CCH® Tagetik | **ACADEMY**

Data integration with DTP



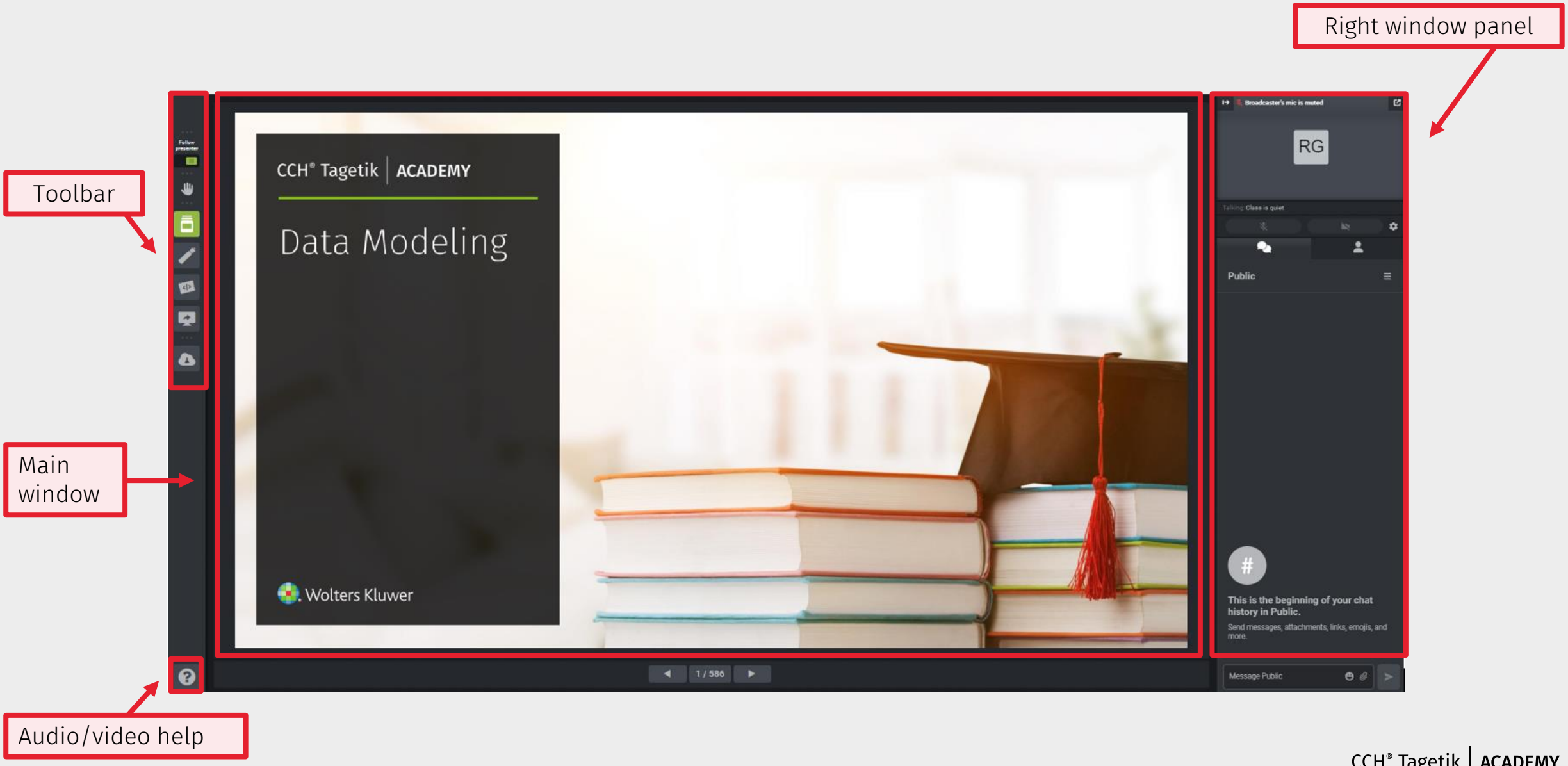
Wolters Kluwer

Before starting

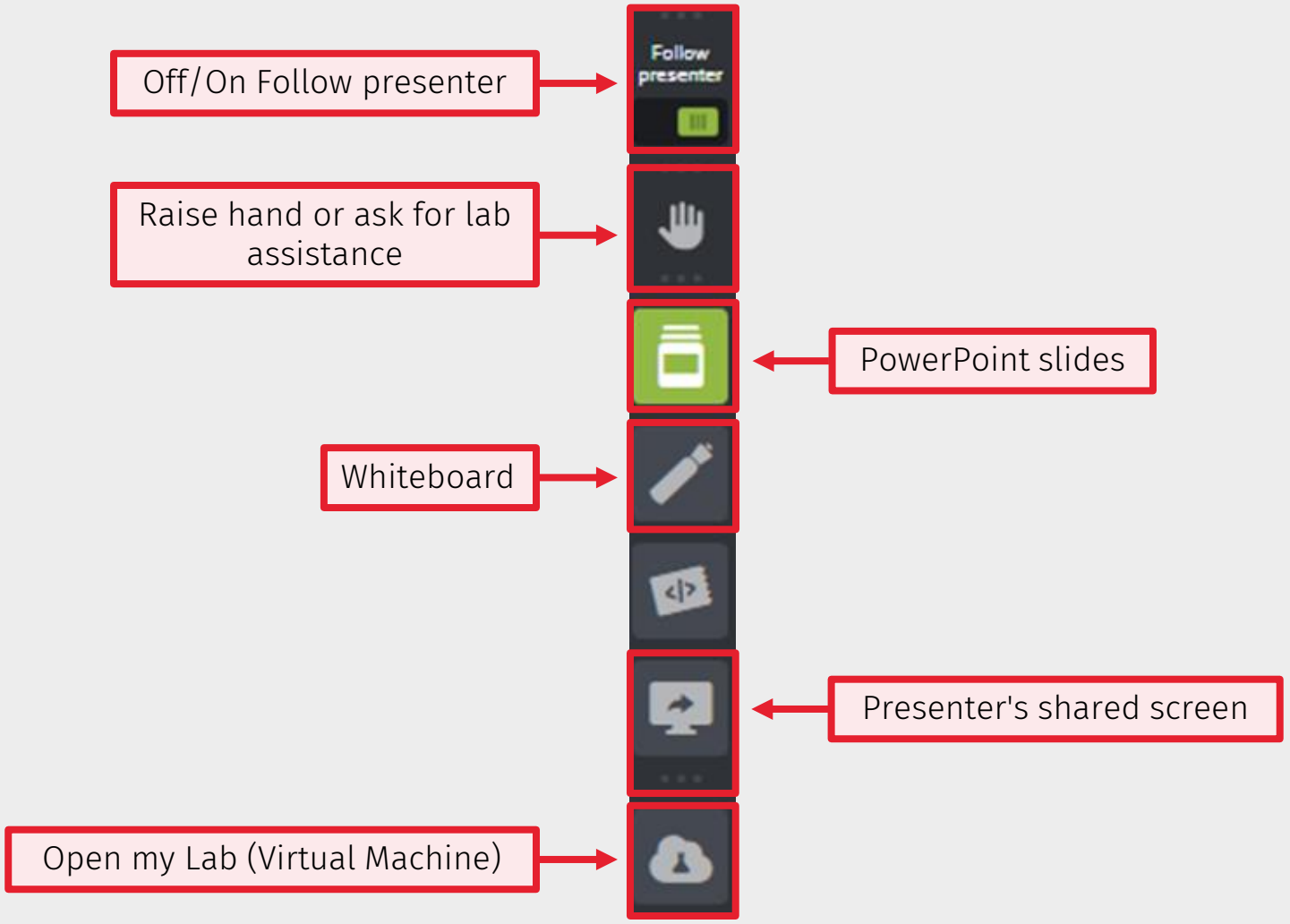
Strigo platform - “How to”



Strigo overview (for Instructor-led classes)



Strigo left toolbar

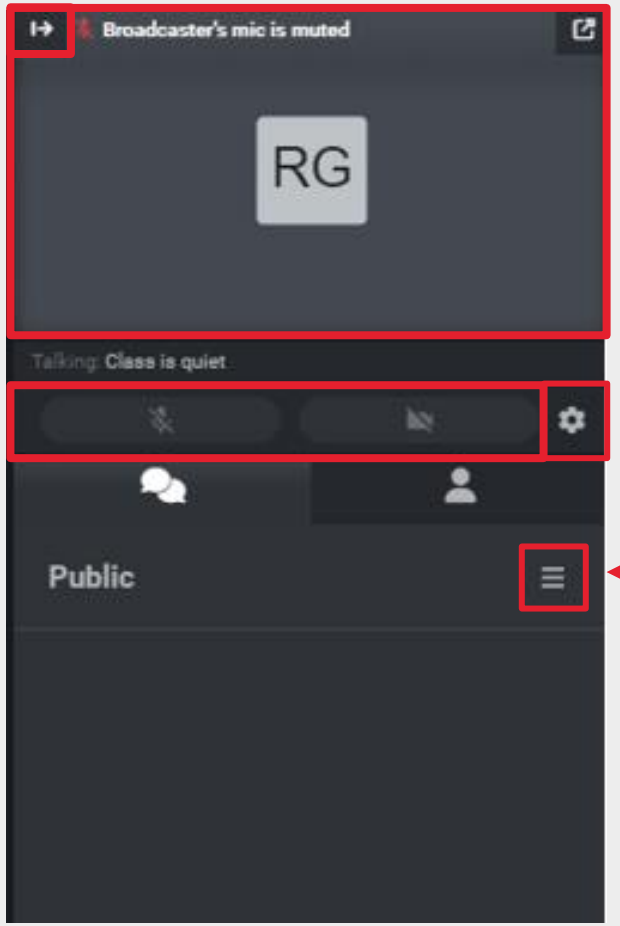


Strigo right window panel

Hide/show window panel

Web cam presenter & who's talking

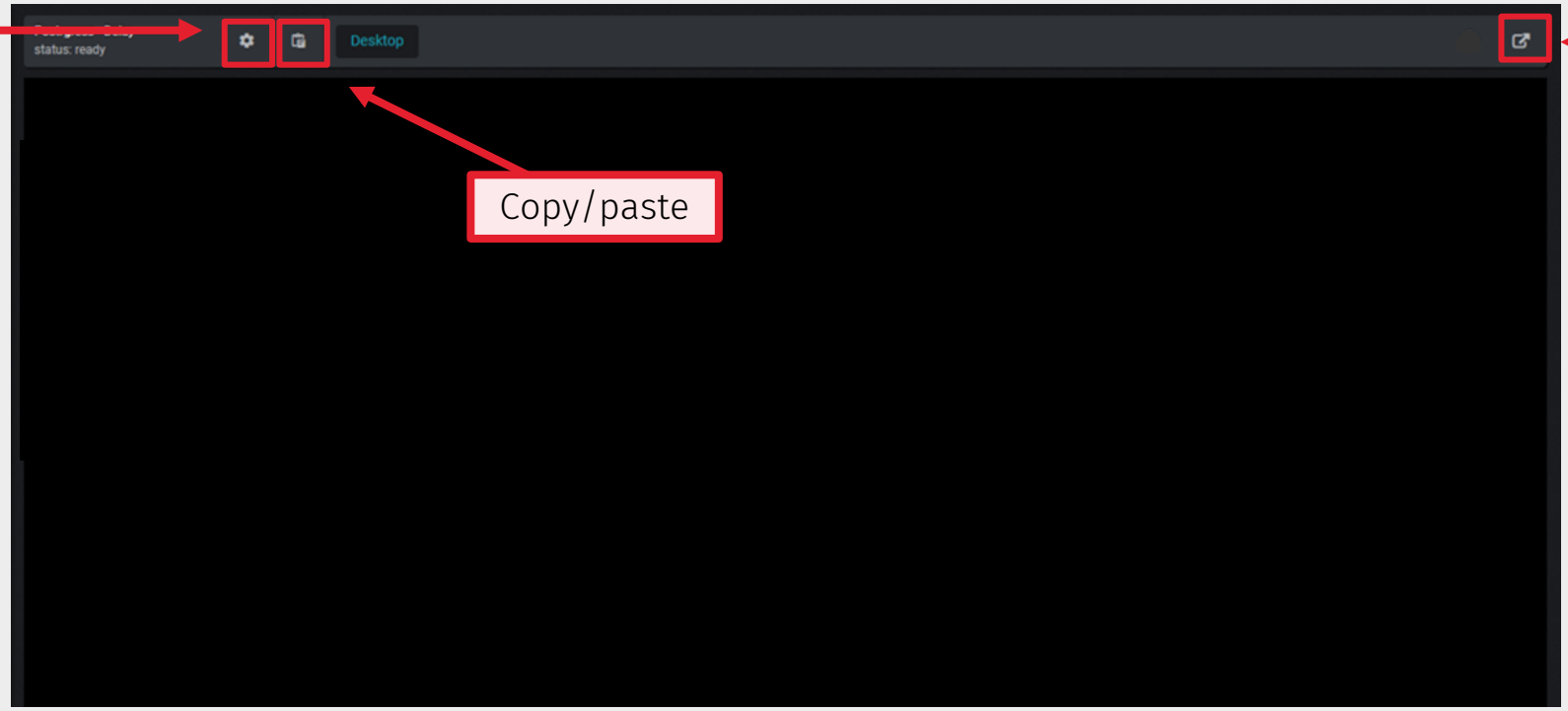
Off/On microphone and camera



A/V Troubleshoot

Open chat menu to switch between public chat and private chat

Strigo Lab (Virtual Machine)



Agenda

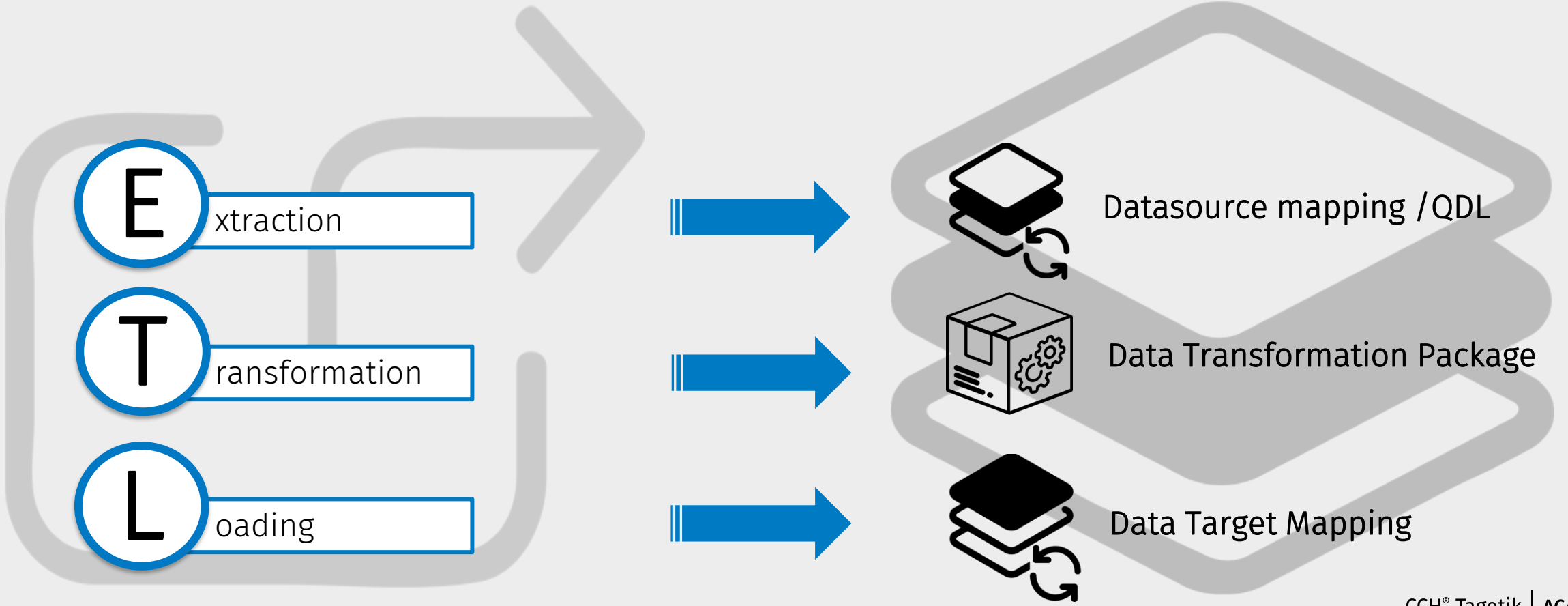


- Reference dataset
- Data source mapping
- Use of dictionaries in AW
- Datasource and virtual dataset
- Parameters
- DTP typical actions on data:
  - Loading
  - Aggregate
  - Unpivot
  - Deploy
  - Calculate
  - Check

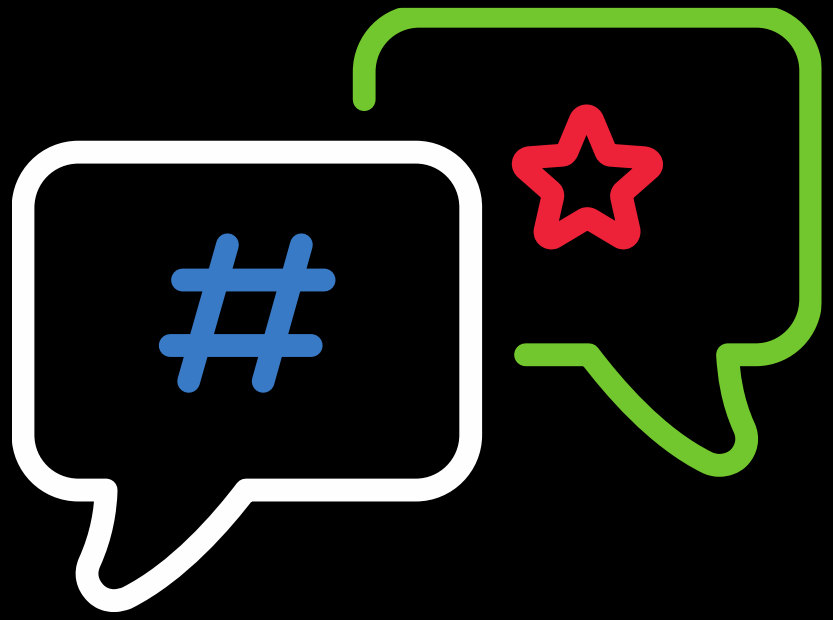
Data Integration in CCH Tagetik platform

FW approach

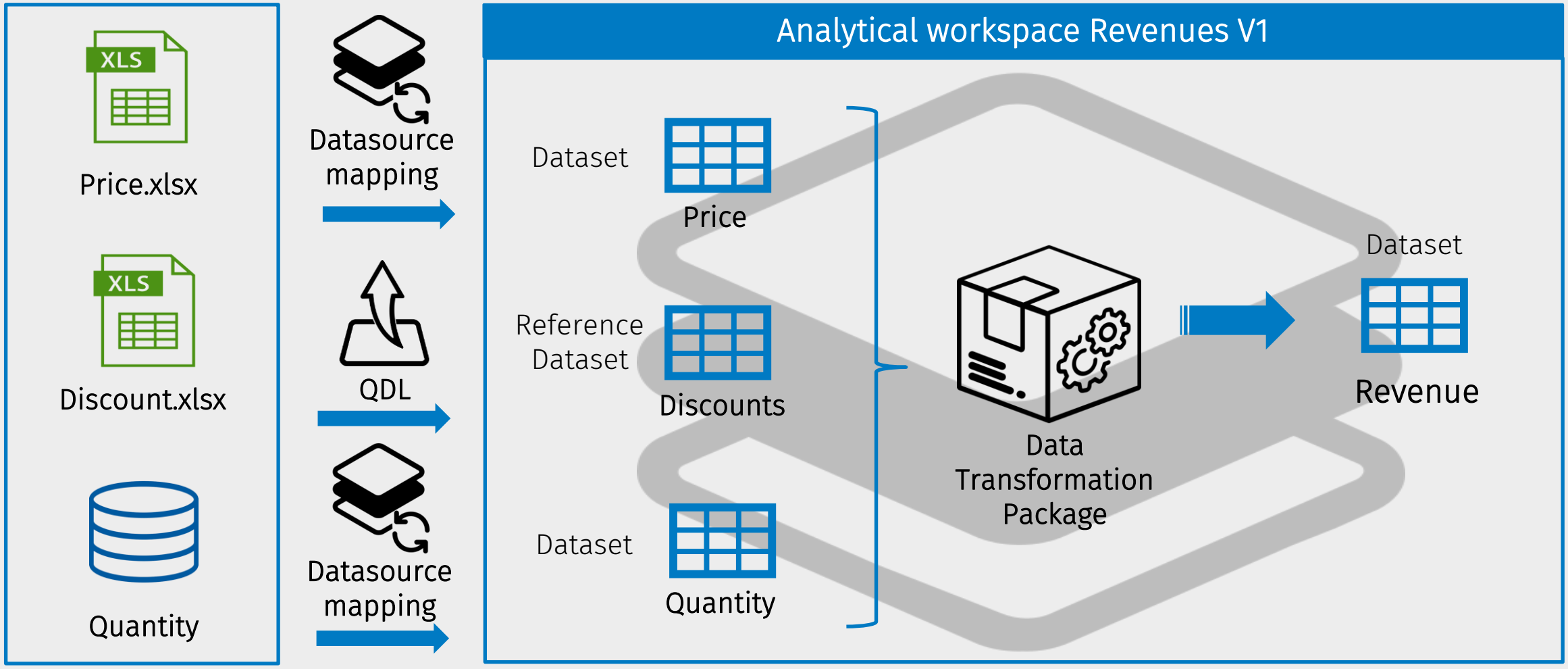
AIH approach



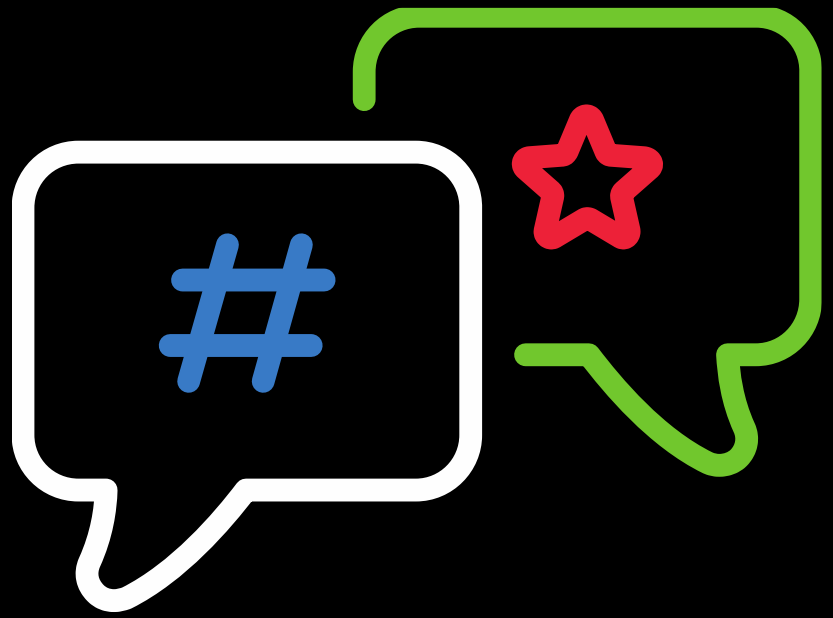
Let’s talk about: Business case



Case study: Revenue calculation



Let’s talk about: Reference dataset (Load discounts)



Reference dataset

- Act as a **Look up table** not to store fact data
- Used to enrich data in DTP with operation **Enrichment with reference dataset**
- Can include **global assumption**
- Input fields **[Condition]** – Output fields [**Attribute**]
- Relationship field [Input: Output]: 1:1, 1:N but in general N input to M output
- **Typologies not available**

Reference dataset – Typical use cases

Dimensions of different sources	
ERP Account (Condition)	Tagetik Account (Attribute)
X01 – Revenue ERP1	501 – Gross Revenues
X02 – Revenue ERP2	501 – Gross Revenues
X03 - Revenue ERP3	502 – Other Revenues

Different level of detail	
SKU (Condition)	PRODUCT (Attribute)
S01 - Norton	P01 -Antivirus
S02 - Avast	P01 - Antivirus
S03 - Ubuntu	P02 – Op. System

Global Assumptions	
SKU Type (Condition)	Discount (Attribute)
SW - Software	5%
HW - Hardware	10%
EL - Electronics	7%

Reference dataset – Example 1:1

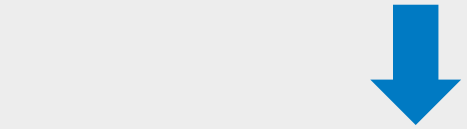
Reference dataset	
SKU (Condition)	PRODUCT (Attribute)
S01 - Norton	P01 -Antivirus
S02 - Avast	P01 - Antivirus
S03 - Ubuntu	P02 – Op. System



Target dataset			
SCENARIO	SKU	PRODUCT	PRICE
2050BDG	S01 - Norton	P01 - Antivirus	100
2050BDG	S02 - Avast	P01 - Antivirus	200
2050BDG	S03 - Ubuntu	P02 – Op. System	300

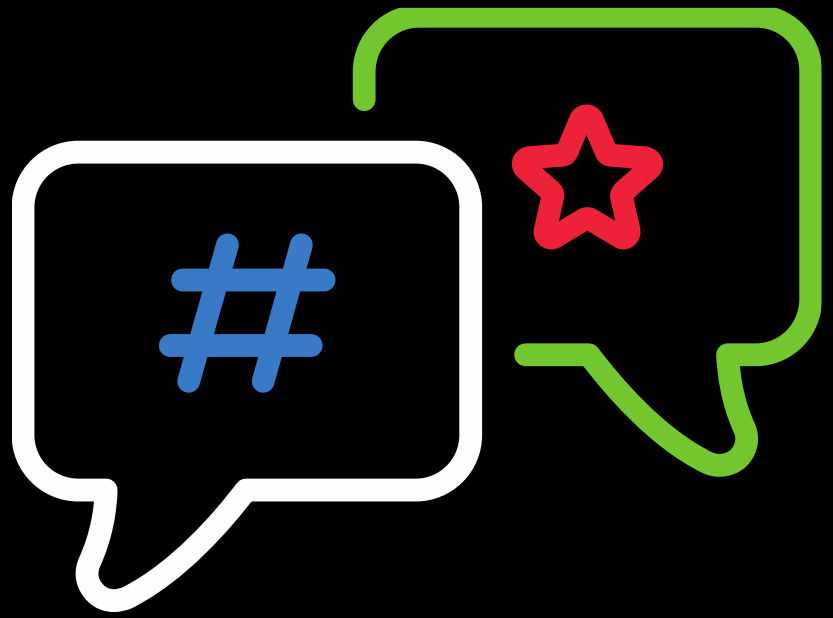
Reference dataset – Example N:1

Reference dataset		
COST CENTER (Condition)	ERP ACCOUNT (Condition)	TGK ACCOUNT (Attribute)
SLS - Sales	X01 – Revenue ERP1	501 –Gross revenues
SLS_UK – Sales UK	X02 – Revenue ERP2	502 –Other revenues
PRD - Production	C03 – Cost ERP1	600- Material costs



Target dataset			
COST CENTER	ERP ACCOUNT	TGK ACCOUNT	AMOUNT
SLS	X01	501 – Gross revenues	100
SLS_UK	X02	502 – Other revenues	200
PRD	C03	600 – Material costs	300

Let’s talk about: Advanced condition using wildcard characters



Ref. dataset – Advanced conditions: wildcard char (%)

SQL like operator **“% “** can be used in condition definition

Example defining **“X% “** means **“Whatever starts with X”**

Reference dataset	
ERP ACCOUNT (Condition)	TGK ACCOUNT (Attribute)
X%	501 – Gross revenues
Y%	601 – Production costs



Target dataset			
COST CENTER	ERP ACCOUNT	TGK ACCOUNT	AMOUNT
SLS	X01	501 – Gross revenues	100
SLS_UK	X02	501 – Gross revenues	350
PRD	Y02	601 – Production costs	200
PRD	Y03	601 – Production costs	300

Ref. dataset – Advanced conditions: wildcard char (null)

Reference dataset		
COST CENTER (Condition)	ERP ACCOUNT (Condition)	TGK ACCOUNT (Attribute)
SLS - Sales	X01 – Revenue ERP1	501 –Gross revenues
SLS_UK – Sales UK	X02 – Revenue ERP2	502 –Other revenues
PRD - Production		600- Production costs



Target dataset			
COST CENTER	ERP ACCOUNT	TGK ACCOUNT	AMOUNT
SLS	X01	501 – Gross revenues	100
PRD	C02	600 – Production costs	200
PRD	C03	600 – Production costs	300

Standard dataset Vs. Reference dataset

	Dataset	Reference dataset
Logical	Fact Table	Lookup table
Technical	• Can be DTP Target • Allows typologies	• Used for enrich data in DTP • Doesn't allow typologies
Commercial	Limited number due to license	No number limitations

Basic Licence: limitation on licence

Number of workspaces

- No limitations on total number of AW

Number of Analytical dimensions

- No limitations

Number of datasets

- The basic AIH licence sets to 15 the number of available datasets
  - **Extensions:** CCH Tagetik can be asked for an extended version of dataset number
  - **Reference Datasets**: they are not included in the list. Can be used with no limitations

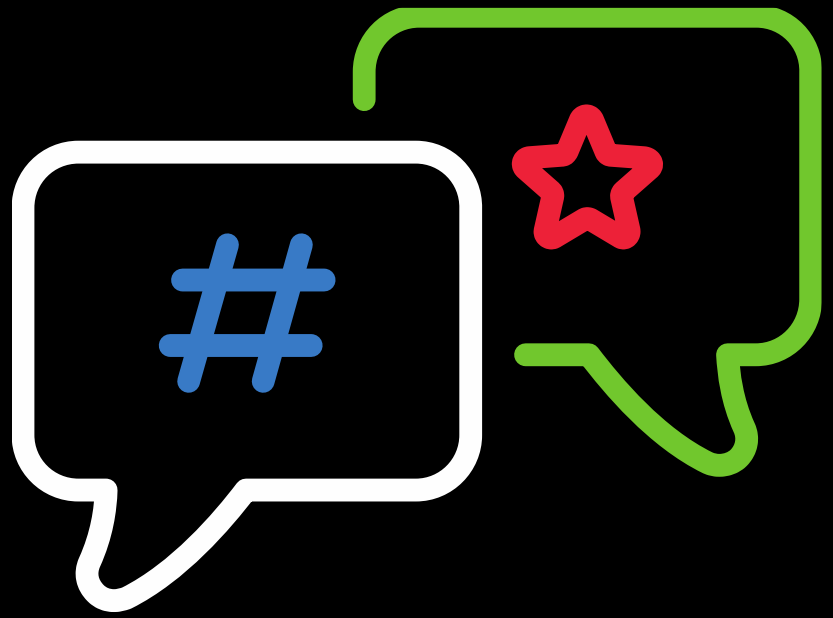
Now it’s your turn

Exercise # 1

Reference dataset definition



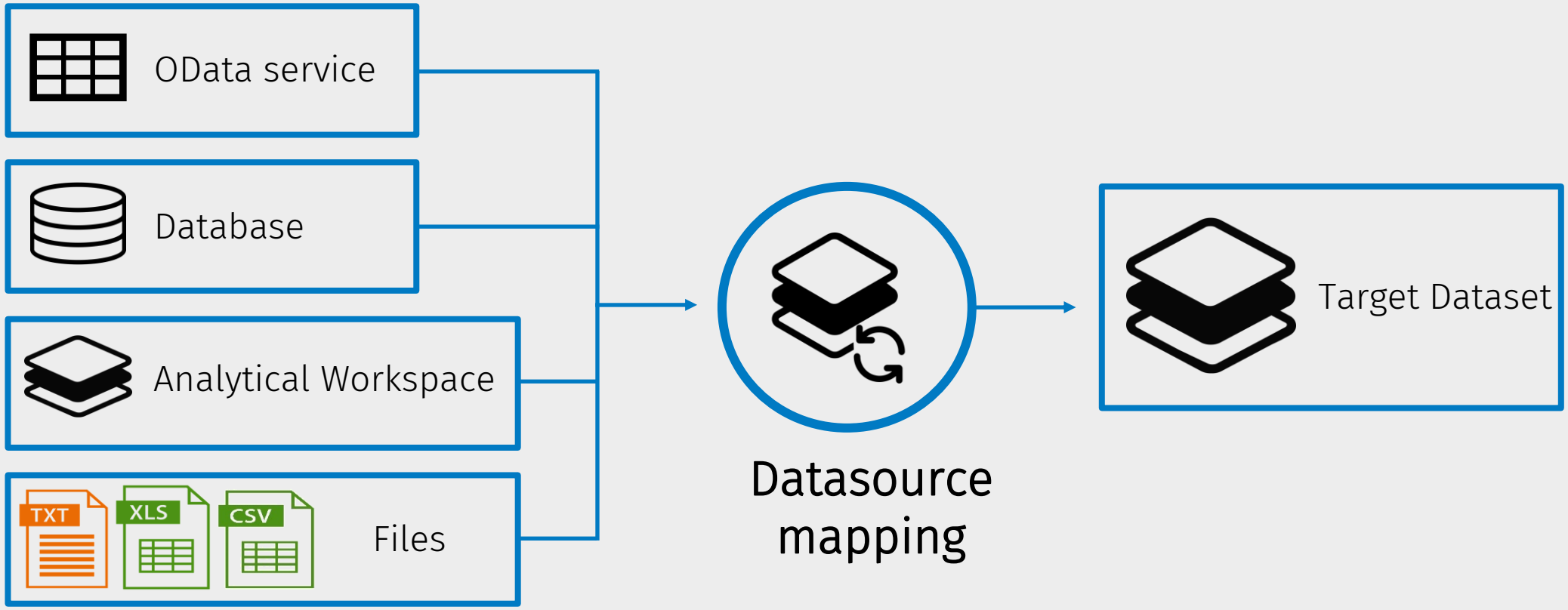
Let’s talk about: Datasource mapping (Load Price list)



Definition Datasource mapping (DSM)

DSM allows you to setup the execution of a Quick Data Loader to load data in a target dataset

Datasource mapping: possible sources



Datasource mapping features

Mapping defined and stored in target dataset structure

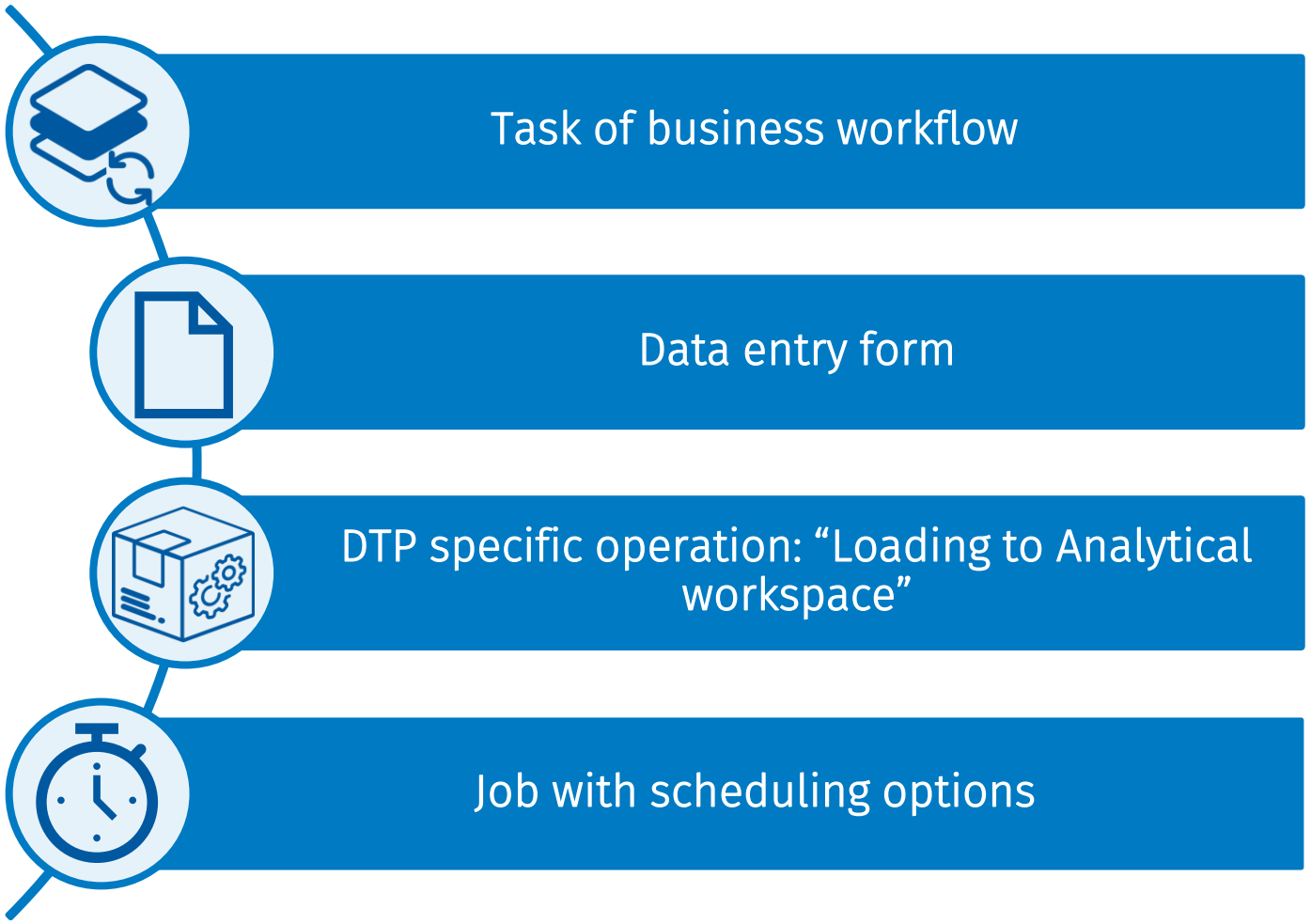
Check user rights and Editability

Can load data on Reporting and Calculation layers

Generates rows with Origin **QDL**

Different deletion policies available

How to run a Datasource mapping



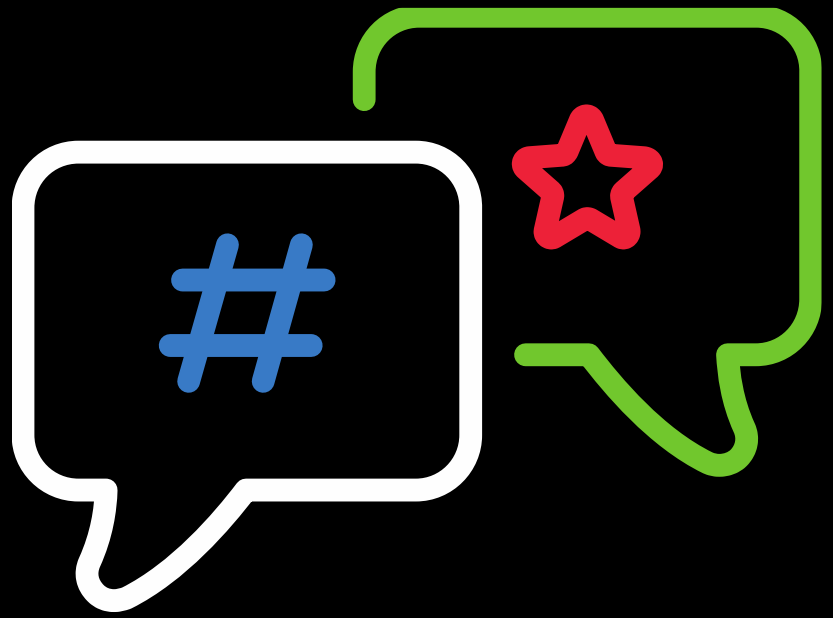
Now it’s your turn

Exercise # 2

Setup and use of Datasource Mapping



Let’s talk about: Use of Dictionaries in AW (Data entry on Quantity)



Use of dictionary in AW

Dictionary can be defined with a query and assigned to specific dataset fields

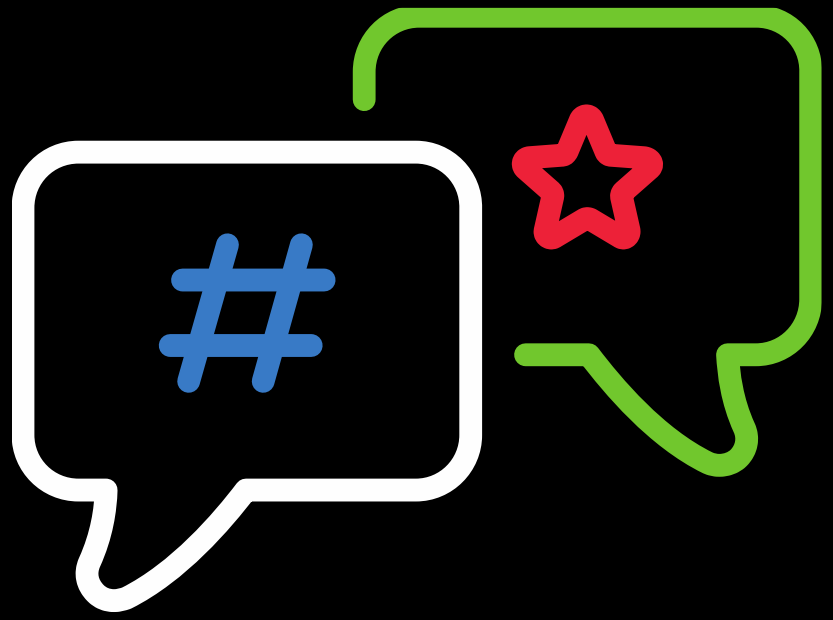
Now it’s your turn

Exercise # 3

Use of Dictionaries in Analytical workspace



Let’s talk about: Datasources (Setup for massive loads of Quantity)



Datasource: main features

Logical definition (View) based on a Query (SQL skills required)

Can read and join any table

Can use Grouping function, sum and any other SQL operation

Retrieved fields can be mapped in CCH Tagetik dimensions and measures

Definition is permanent, data retrieved depend on the moment of access

Result affected by user rights

Parameters available

Datasource typical uses

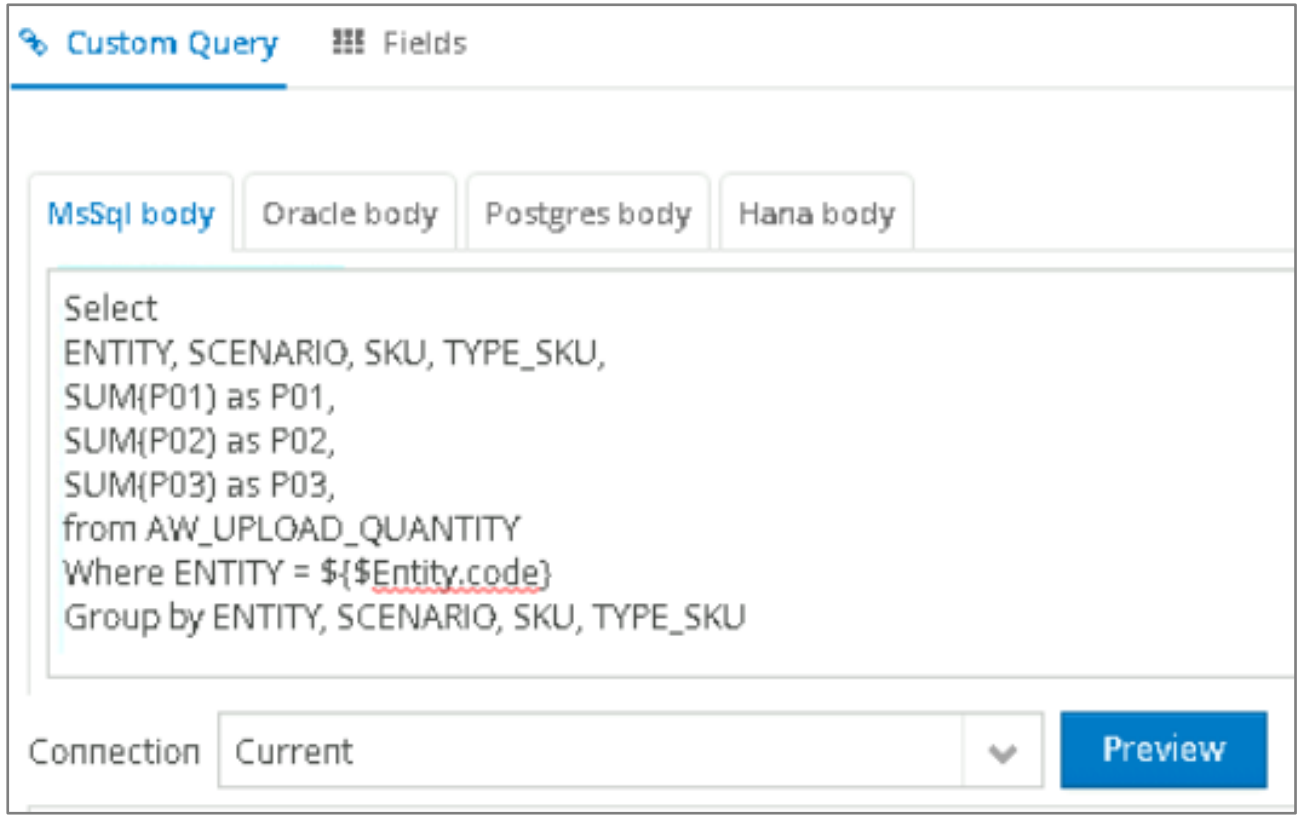
DTP & Datasource mapping

- Data extraction
- Data preparation (preliminary operations)
- Source for some DTP operations
- Source for Datasource mapping

Reporting

- Creation of forms that hold data not retrievable from datasets
- Multidimensional matrices if Scenario and Period are present in structure

Datasource - Example



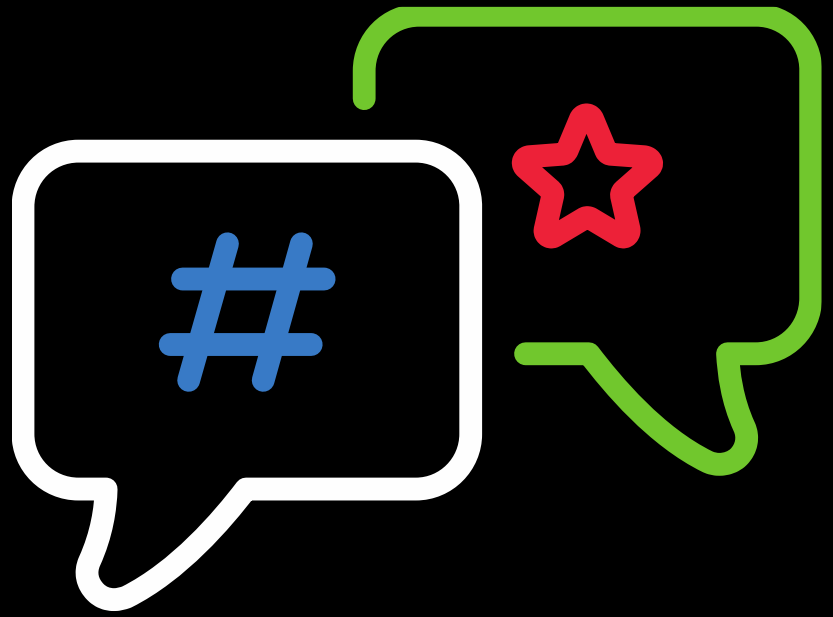
Now it’s your turn

Exercise # 4

Datasource definition



Let’s talk about: DTP



Data Transformation Package (DTP)

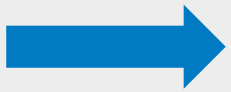
- Calculation engine of AW
- Data processing that can include many calculation activities
- Result of each activity is stored on a target dataset
- Generates rows with Origin DTP_ELAB
- Can be run in a task list, form, cockpit, job

DTP - Activity

Set of operations that acts on the same target dataset

Activity 1: Sales Revenue IT

A) Insert from dataset
B) Join from another dataset
C) Enrichment with Formula



Revenues (Typology Sales IT)			
Scenario	Period	Product	Amount
2050BDG	06	P01	200
2050BDG	09	P02	350
2050BDG	12	P03	170

Result is loaded to a target dataset in a specific partition called **Typology**

Data Transformation Package - Operations

Insertion operations

- Add records in a dataset

Enrichment operations

- Update or populate columns into non-empty datasets

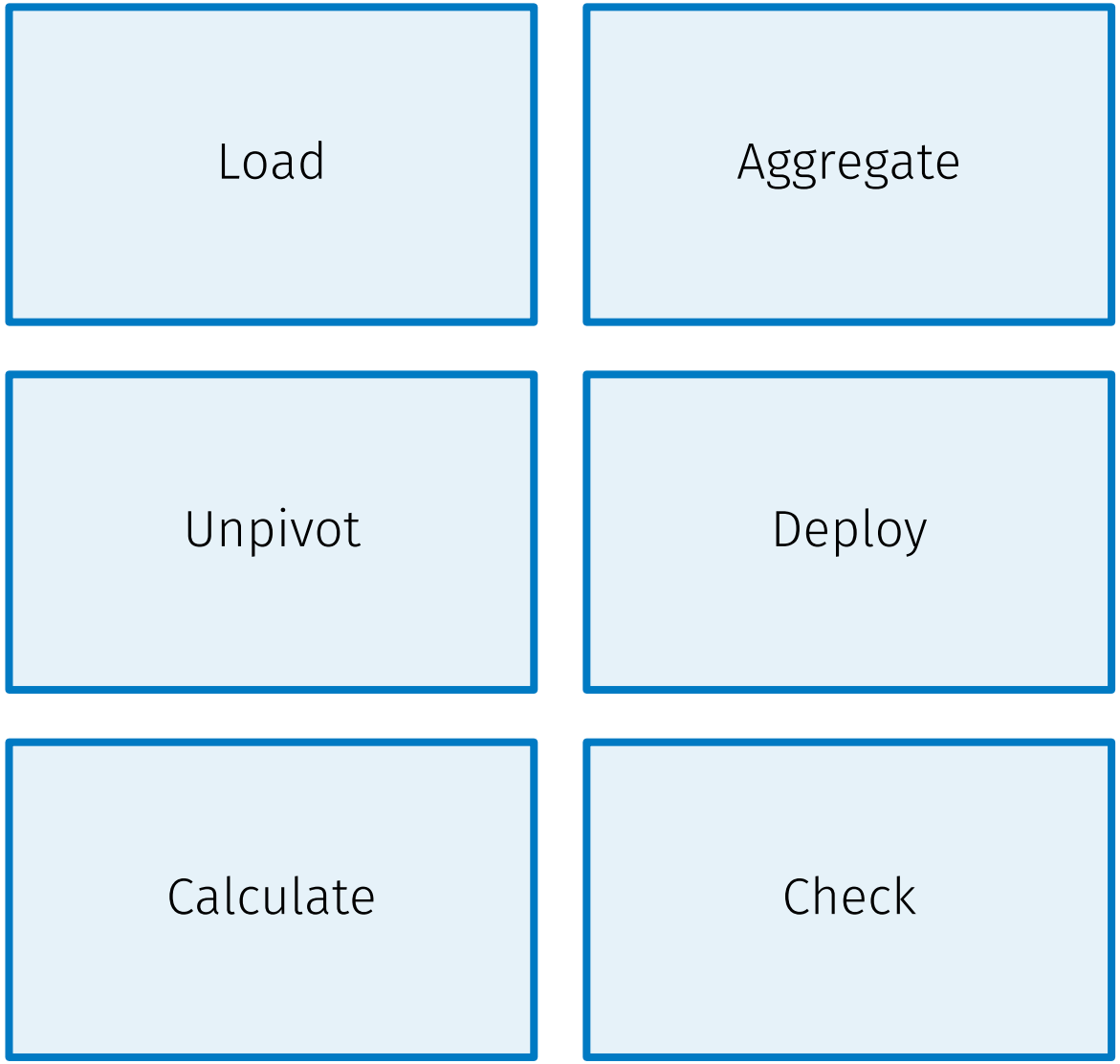
Check operations

- Verify consistency of the values

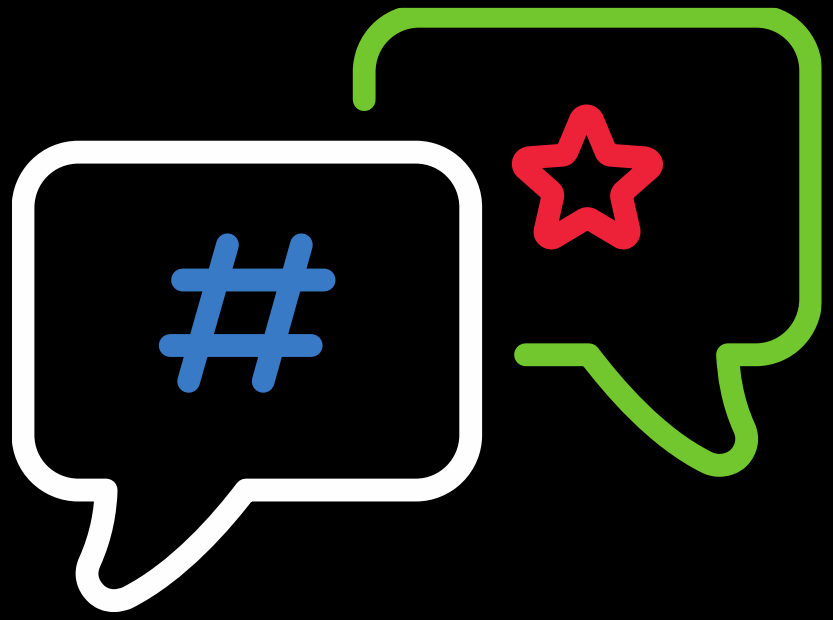
Special methods

- Custom script (stored procedure) enriched with intelligent parameters

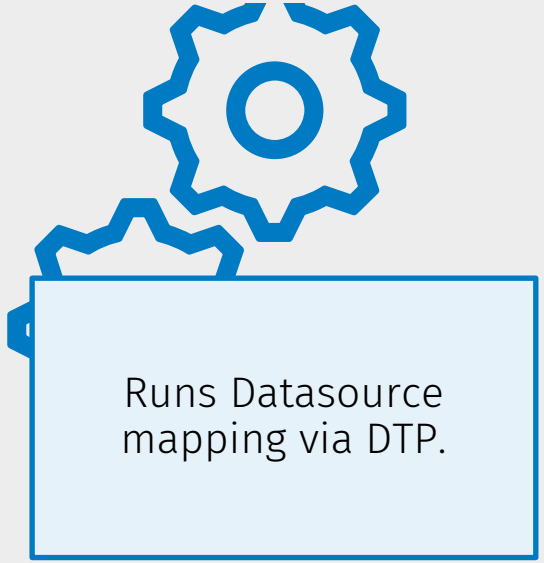
DTP typical actions on data



Let’s talk about: Loading Activities – Operation Loading to Analytical Worspace (massive loads of Quantity)



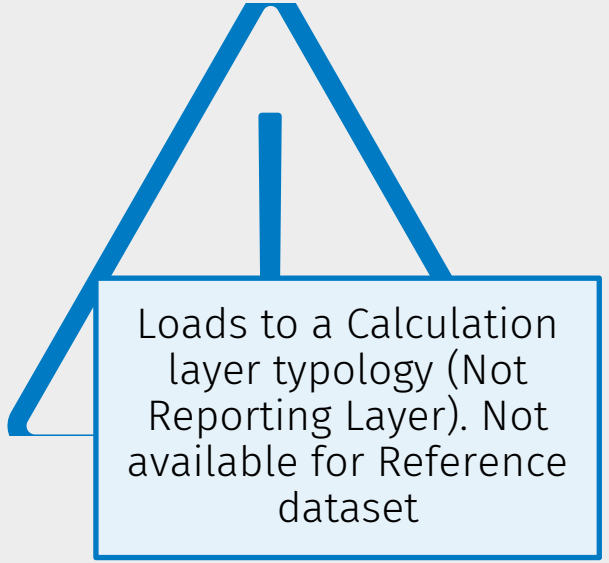
Loading to Analytical Workspace



Runs Datasource mapping via DTP.



Allows you to include Datasource mapping running into DTP along with other activities



Loads to a Calculation layer typology (Not Reporting Layer). Not available for Reference dataset

Loading to Analytical Workspace

Allows you to run Datasource mapping along with other DTP activities in a single task



Loading to Analytical Workspace

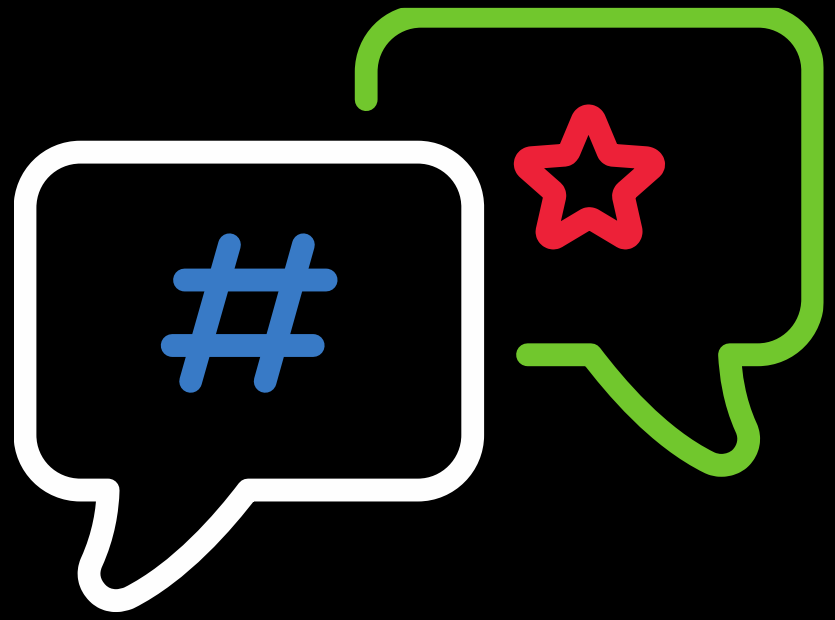
Do's

- Extraction from:
  - OData service
  - Database
  - Analytical Workspace
  - Files on server

Don'ts

- Extraction from Local file
- Limitations in case of DTP run with Execution independent from context

Let’s talk about: Pro tips: use of Reporting layer filter



Reporting layer filter - Activity

If **Update reporting layer** option is enabled, the system, at the end of the activity, performs these actions:

- Delete data on Reporting layer
- Update Reporting layer with result held in Calculation layer



What if I want to avoid the deletion of some data on Reporting layer while updating?

Use Reporting layer filter

Reporting layer filter - Activity

SQL instructions when DTP Activity performs **Update Reporting Layer :**

Reporting layer filter empty

```
DELETE FROM AW_001_000001_000001

INSERT INTO AW_001_000001_000001…..
```

Delete all data from Reporting layer

Reporting layer filter defined

```
DELETE FROM AW_001_000001_000001
WHERE ('REPORTING LAYER FILTER')

INSERT INTO AW_001_000001_000001…..
```

Delete data according to Reporting layer filter condition

Now it’s your turn

Exercise # 5

Loading data :
Pro tips

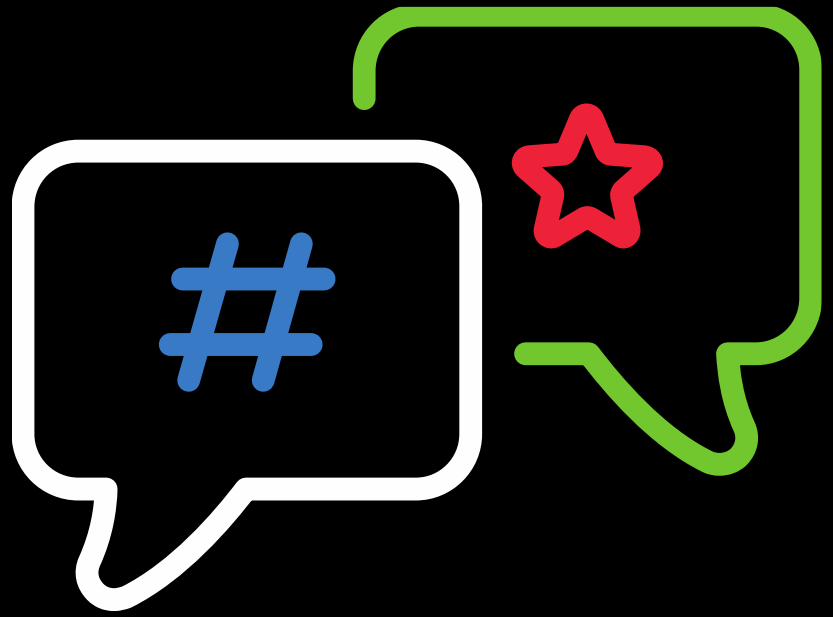


Loading to Analytical Workspace: Origin of data

	Origin of loaded data using “Loading to AW”
Calculation layer	QDL
Reporting Layer	DTP_ELAB

Origin field, keep track of the use of «Loading to AW» on Calculation layer

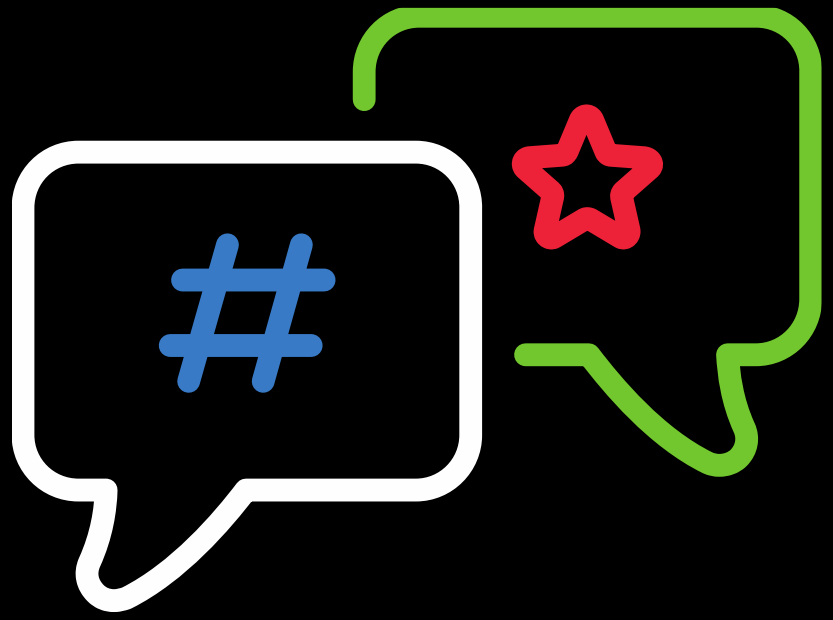
Let’s talk about: DTP case study



Data Transformation Package: List of activities

D.T.P.	
Activity 0	Loading
Activity 1-2-3	Data preparation
Activity 4	Revenue calculation

Let’s talk about: DTP Multiple steps of calculation



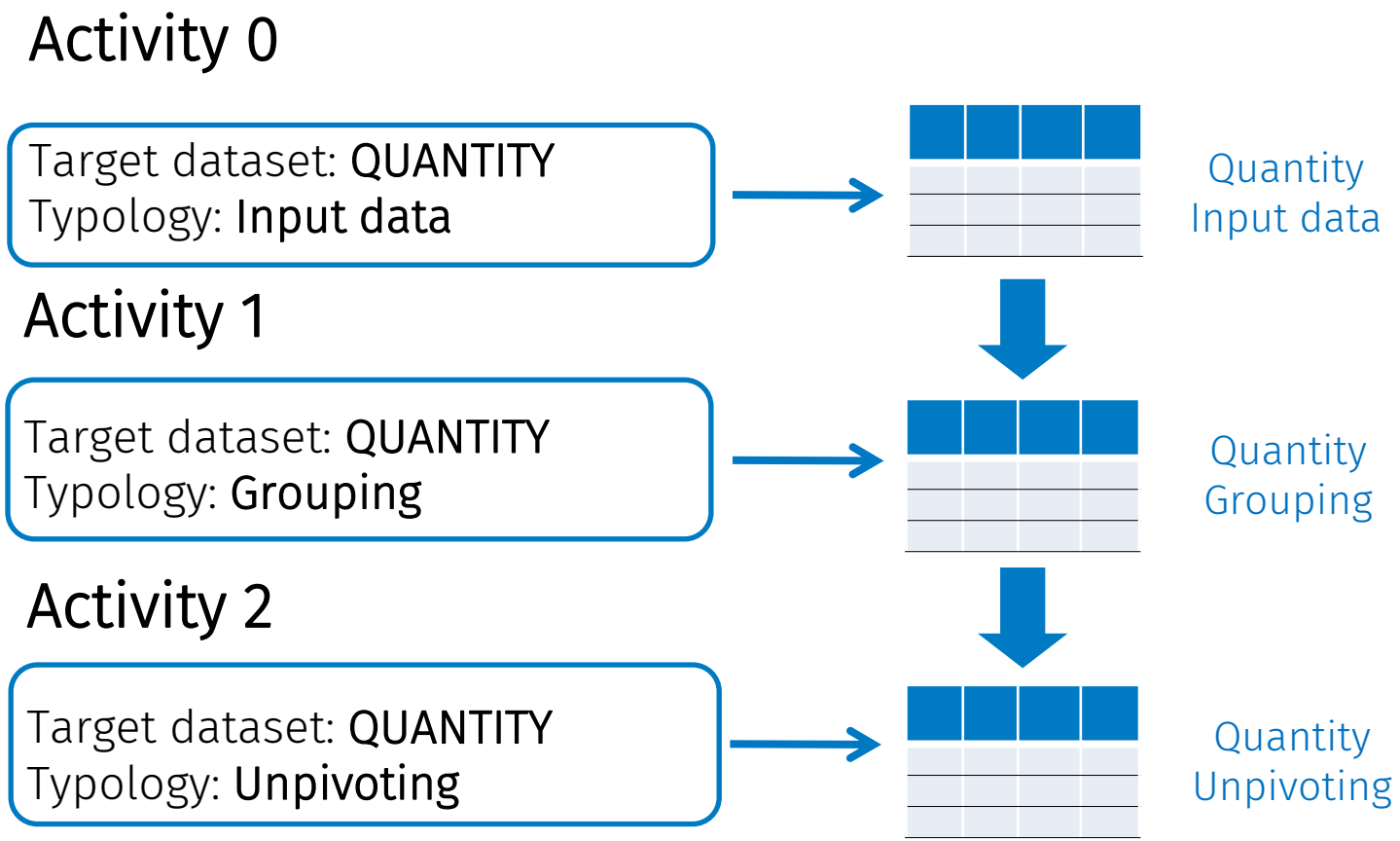
Multiple steps of calculation in DTP: features

Same target dataset, Different typologies

Previous target typology is the input of following activity

Different activities represent sequence of calculation

Multiple steps of calculation in DTP: example



Multiple steps of calculation: Why?

Freeze partial result in a typology to be input for other calculations

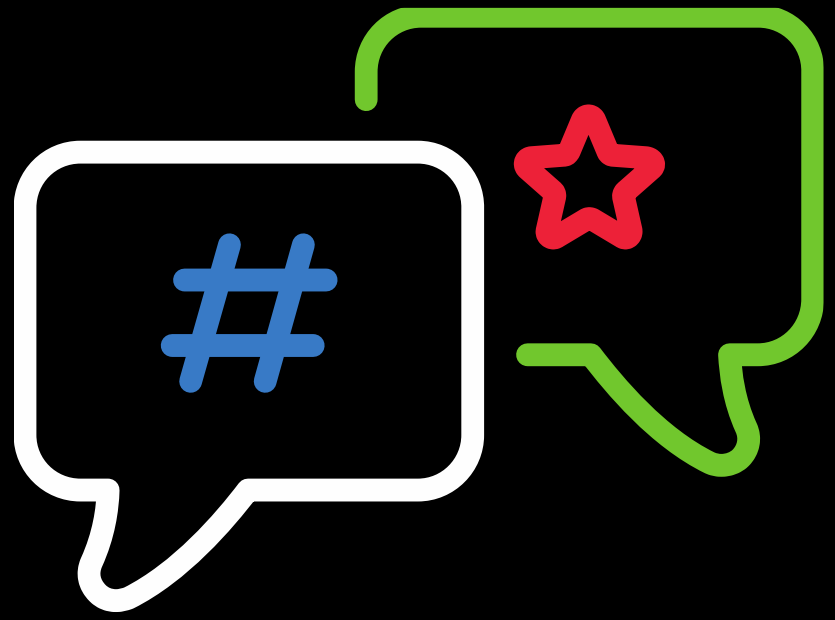
Incoming requirement while part of calculation has been already performed

Calculation needs

Partialize difficult calculation in different steps

Audit and logging

Let’s talk about: Insert operation refresh



Insert from another dataset

Source		
SCENARIO	PERIOD	REVENUE
2050FCT	06	1000
2051BDG	12	2000



Target			
SCENARIO	PERIOD	REVENUE	PRODUCT
2050FCT	06	1000	
2051BDG	12	2000	

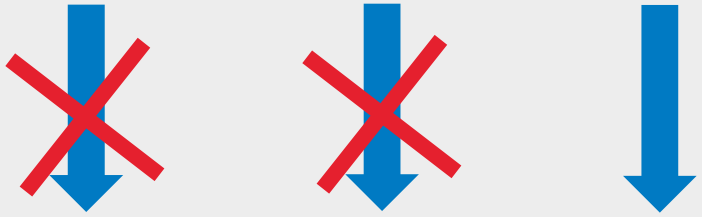
Mostly used to fill empty dataset

Stores Source records into Target Dataset depending on matching fields

Matching fields: Definition of relationships between Source fields and Target fields

Insert from another dataset

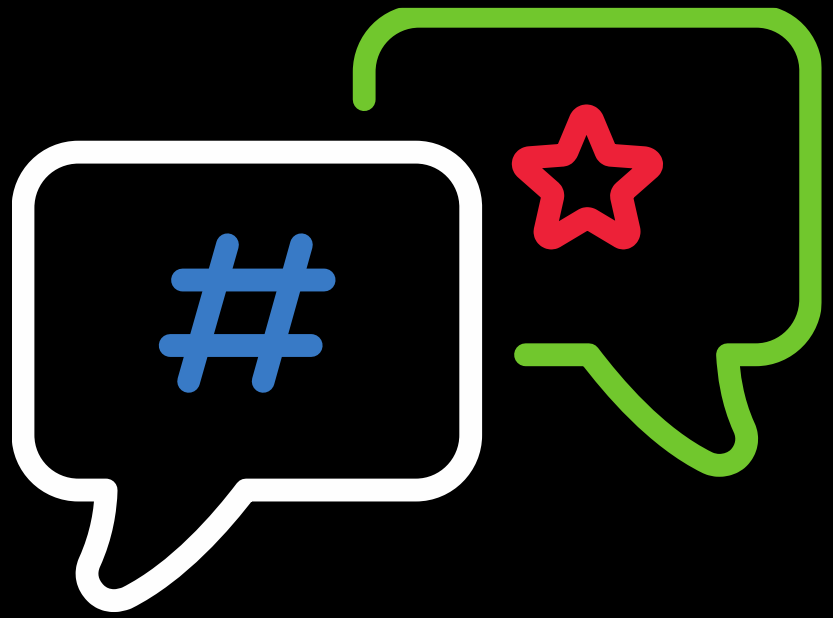
Source		
SCENARIO	PERIOD	REVENUE
2050FCT	06	1000
2051BDG	12	2000



Target				
SCE	PER	PRICE	ENTITY	PRODUCT
		1000		
		2000		

If matching fields is missing, system loads new rows on target dataset filling only fields whose codes match

Let’s talk about: DTP: How to aggregate data



Grouping functions

Definition

- In the matching fields window of some operations

Grouping mode

- Dimensional fields: **Group by**
- Numeric fields: **SUM, COUNT, AVERAGE, MAX,MIN**

Why

- Reduce number of rows to be processed (improve performance)
- Calculation needs

Grouping function: reduce rows example

Before Grouping

Group by	Group by	Sum
SCENARIO	PRODUCT	QUANTITY
2051BDG	Mouse	200
2051BDG	Mouse	300
2052BDG	Laptop	100
2052BDG	Laptop	500

Grouping function: reduce rows example

After Grouping

Group by SCENARIO	Group by PRODUCT	Sum QUANTITY
2051BDG	Mouse	500
2052BDG	Laptop	600

- Elements grouped by Scenario and Product
- Quantity is the sum of the rows grouped

Grouping function: Calculation needs example

Group amounts that are in different rows in a single row

Group by	Group by	Sum	Sum	Sum	Sum
SCENARIO	PRODUCT	Revenue Q1	Revenue Q2	Revenue Q3	Revenue Q4
2051BDG	Mouse	200			
2051BDG	Mouse		300		
2051BDG	Mouse			100	
2051BDG	Mouse				500

Grouping function: Calculation needs example

Group by	Group by	Sum	Sum	Sum	Sum
SCENARIO	PRODUCT	Revenue Q1	Revenue Q2	Revenue Q3	Revenue Q4
2051BDG	Mouse	200	300	100	500

Calculation can go on with next step

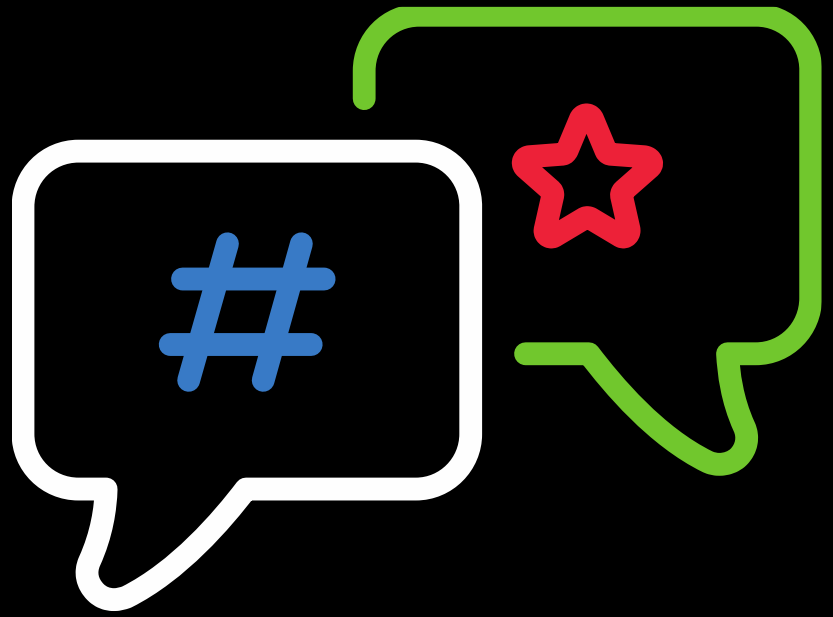
Now it’s your turn

Exercise # 6

Aggregate data with grouping functions



Let’s talk about: Unpivoting data



Revenue calculation: Input and target dataset

Input

Quantity							
Scenario	Entity	SKU	SKU Type	P01	P02	....	P12
2053BDG	B00	S001	SW	10	20		120

Price				
Scenario	Period	Entity	SKU	Price
2053BDG	01	B00	S001	100
2053BDG	02	B00	S001	95
2053BDG	12	B00	S001	98

Target

Revenue							
Scenario	Period	Entity	SKU	SKU Type	Price	Quantity	Revenue

Revenue calculation: Input and target dataset

Input

Target

Quantity

Scenario	Entity	SKU	SKU Type	P01	P02	....	P12
2053BDG	B00	S001	SW	10	20		120

Price

Scenario	Period	Entity	SKU	Price
2053BDG	01	B00	S001	100
2053BDG	02	B00	S001	95
2053BDG	12	B00	S001	98

Insert from another dataset

Revenue

Scenario	Period	Entity	SKU	SKU Type	Price	Quantity	Revenue

Revenue calculation: Input and target dataset

How can I load Quantity on target? Dataset structure is different

Input

Quantity							
Scenario	Entity	SKU	SKU Type	P01	P02	....	P12
2053BDG	B00	S001	SW	10	20		120

Target

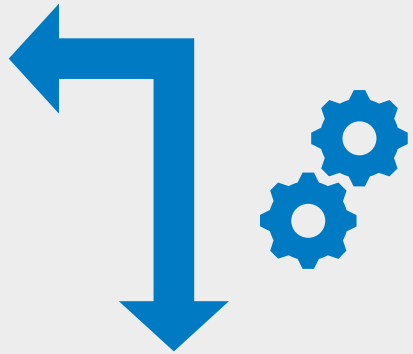
Revenue							
Scenario	Period	Entity	SKU	SKU Type	Price	Quantity	Revenue
2053BDG	01	B00	S001		100		
2053BDG	02	B00	S001		95		
2053BDG	12	B00	S001		98		

Unpivoting

It is necessary to unpivot the **Quantity** dataset

Quantity and Unpivot Quantity

Quantity							
Scenario	Entity	SKU	SKU Type	P01	P02	....	P12
2053BDG	B00	S001	SW	10	20		120



Unpivot Quantity					
Scenario	Entity	SKU	SKU Type	Period	Quantity
2053BDG	B00	S001	SW	01	10
2053BDG	B00	S001	SW	02	20
2053BDG	B00	S001	SW	....	....
2053BDG	B00	S001	SW	12	120

Deploying rows

1 row in **Quantity** dataset

Generates

12 rows in **Unpivot Quantity** dataset



Calculation needs deploying rows

Quantity structure – Technical fields necessary for deployment

- **Period (dimension field)** : value fixed to 12 in each row
- **Period Pivot (Text field)** : Technical field used to hold real value of Period after deployment
- **Quantity:** Number field to hold the unpivoted value of Quantity

Quantity before unpivoting										
Scenario	Entity	SKU	SKU Type	Period	Period pivot	Quantity	P01	P02	....	P12
2053BDG	B00	S001	SW	12			10	20		120

Quantity after unpivoting										
Scenario	Entity	SKU	SKU Type	Period	Period pivot	Quantity	P01	P02	....	P12
2053BDG	B00	S001	SW	12	01	10	10	20		120
2053BDG	B00	S001	SW	12	02	20	10	20		120
2053BDG	B00	S001	SW	12	....	....	10	20		120
2053BDG	B00	S001	SW	12	12	120	10	20		120

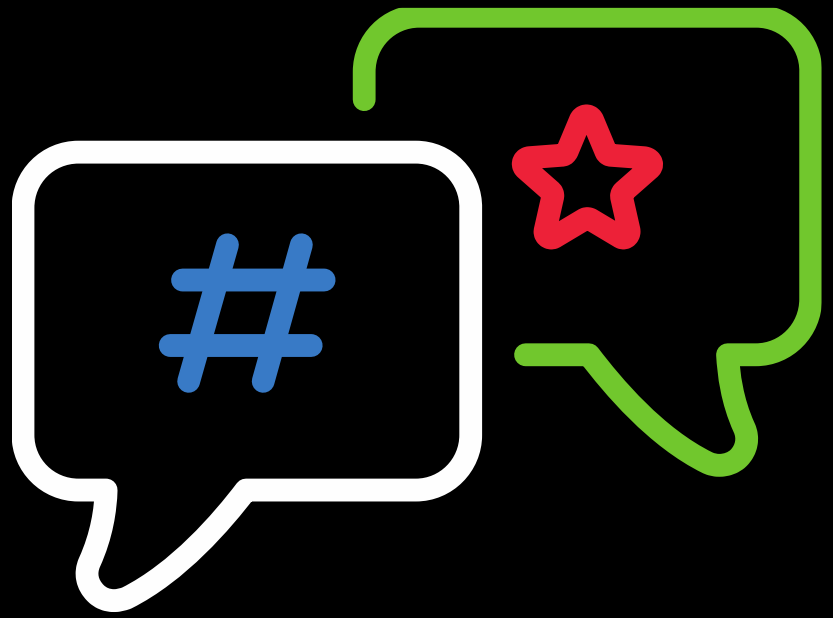
DTP operations

Which operations must the DTP do?

1. Deploy rows

2. Calculate Quantity field with formula

Let’s talk about: Opening rows: Reference dataset - Deploy



Reference dataset - Deploy

- If Look up has multiple relationships
- Enrich and open new rows in target dataset
- Run by “Enrichment with reference dataset”

Reference dataset - deploy

Reference dataset deploy	
Cost center (Condition)	Employee (Attribute)
SLS - Sales	E01 – Adrian Smith
SLS - Sales	E02 – David Coverdale
MRK - Marketing	E03 – Steve Harris
MRK - Marketing	E04 – Jimmy Page
PRD - Production	E05 - John Lord
PRD - Production	E06 – Geezer Butler



Target dataset		
Cost center	Salary increase	Employee
SLS - Sales	0,05	E01 – Adrian Smith
SLS - Sales	0,05	E02 – David Coverdale
MRK - Marketing	0,07	E03 – Steve Harris
MRK - Marketing	0,07	E04 – Jimmy Page
PRD - Production	0,04	E05 – John Lord
PRD - Production	0,04	E06 – Geezer Butler

Reference dataset - deploy

Reference dataset deploy		
PROCESS (Condition)	SCENARIO (Attribute)	PERIOD (Attribute)
2020BUDGET	2020BDG	03 - March
2020BUDGET	2020BDG	06 - June
2020BUDGET	2020BDG	09 - September
2020BUDGET	2020BDG	12 - December



Target dataset			
Process	Expenses	Scenario	Period
2020BUDGET	10000	2020BDG	03 - March
2020BUDGET	10000	2020BDG	06 - June
2020BUDGET	10000	2020BDG	09 - September
2020BUDGET	10000	2020BDG	12 - December

CASE STUDY: Reference dataset – deploy

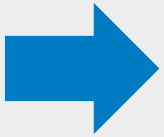
Reference dataset deploy	
PERIOD (Condition)	PERIOD PIVOT (Attribute)
12	01
12	02
12	.....
12	12



Target dataset										
Scenario	Entity	SKU	SKU Type	Period	Period pivot	Quantity	P01	P02	....	P12
2053BDG	B00	S001	SW	12	01		10	20		120
2053BDG	B00	S001	SW	12	02		10	20		120
2053BDG	B00	S001	SW	12	....	....	10	20		120
2053BDG	B00	S001	SW	12	12		10	20		120

Enrichment with formula – Quantity field

Quantity



```
CASE PERIOD_PVT
WHEN '01' THEN P01
WHEN '02' THEN P02
WHEN '03' THEN P03
.....
WHEN '12' THEN P12
ELSE 0 END
```

Scenario	Entity	SKU	SKU Type	Period	PERIOD_PVT	Quantity	P01	P02	....	P12
2053BDG	B00	S001	SW	12	01	10	10	20		120
2053BDG	B00	S001	SW	12	02	20	10	20		120
2053BDG	B00	S001	SW	12	....	....	10	20		120
2053BDG	B00	S001	SW	12	12	120	10	20		120

Unpivot quantity dataset

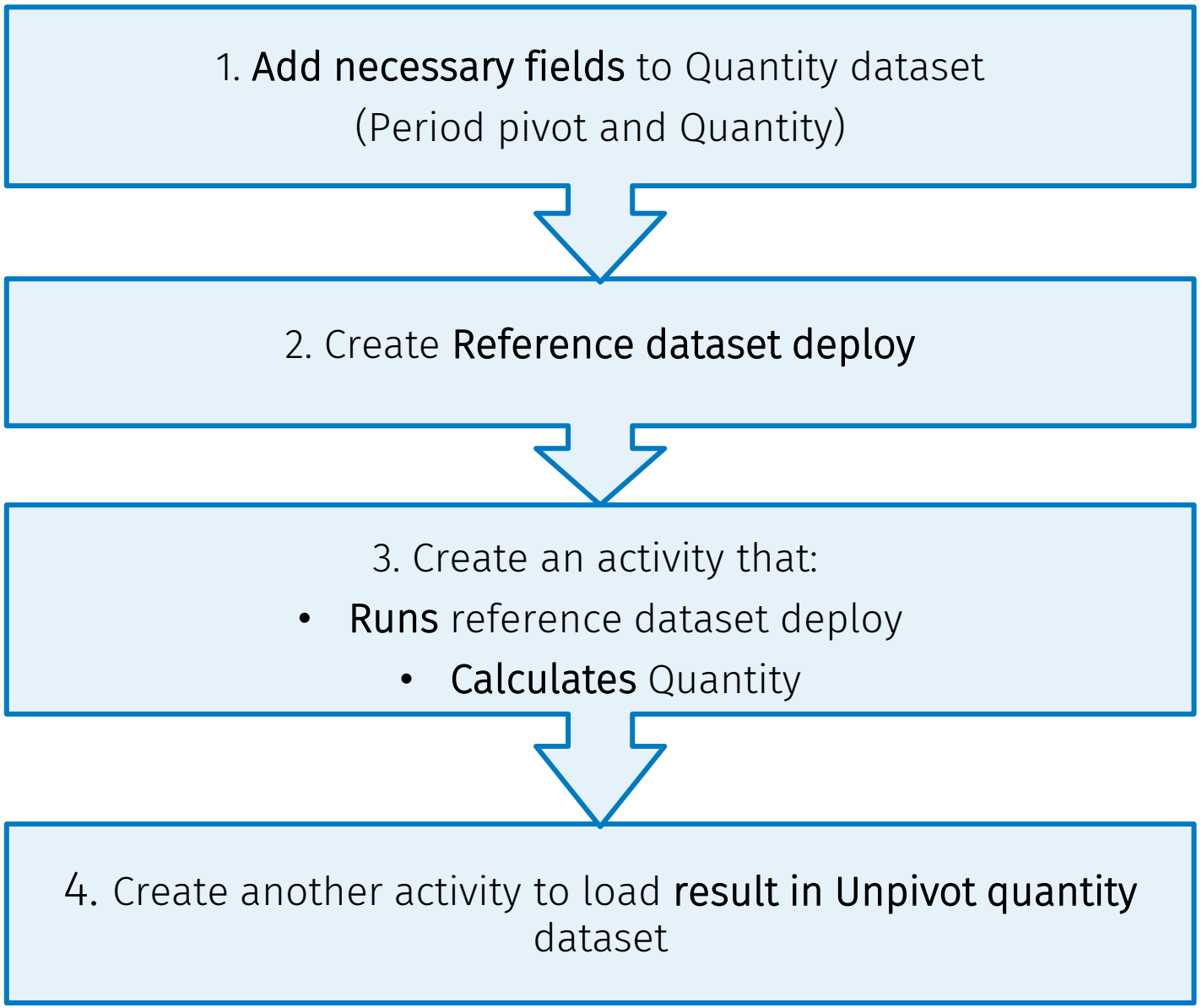
Result is loaded in a target dataset that has only necessary fields for Revenue calculation

Quantity										
Scenario	Entity	SKU	SKU Type	Period	PERIOD_PVT	Quantity	P01	P02	....	P12
2053BDG	B00	S001	SW	12	01	10	10	20		120
2053BDG	B00	S001	SW	12	02	20	10	20		120
2053BDG	B00	S001	SW	12	....	....	10	20		120
2053BDG	B00	S001	SW	12	12	120	10	20		120

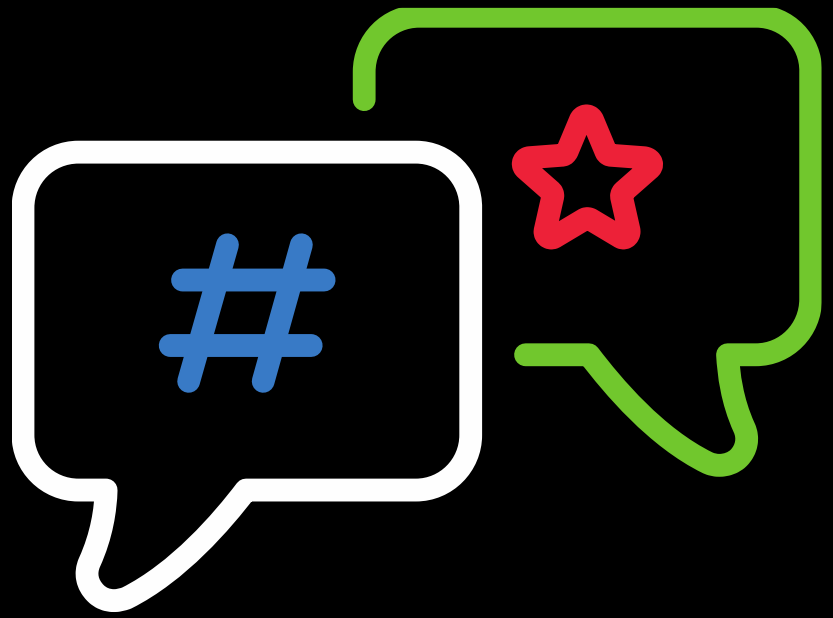


Unpivot Quantity					
Scenario	Entity	SKU	SKU Type	Period	Quantity
2053BDG	B00	S001	SW	01	10
2053BDG	B00	S001	SW	02	20
2053BDG	B00	S001	SW	....	....
2053BDG	B00	S001	SW	12	120

Unpivoting recap



Let’s talk about: Deploying rows: Join from another dataset (refresh)



Join from another dataset

Source			
SCENARIO (Condition)	PRODUCT (Condition)	PERIOD (Attribute)	QUANTITY (Attribute)
2051BDG	Mouse	06	25
2051BDG	Mouse	07	35
2051BDG	Mouse	08	15

Target				
SCENARIO	PRODUCT	PERIOD	QUANTITY	PRICE
2051BDG	Mouse			25

- Open new rows
- Enrich existing columns of a dataset
- **Conditions** fields are the key to perform the join
- **Attributes** fields are the information used to enrich the target

Join from another dataset

Source			
SCENARIO (Condition)	PRODUCT (Condition)	PERIOD (Attribute)	QUANTITY (Attribute)
2051BDG	Mouse	06	25
2051BDG	Mouse	07	35
2051BDG	Mouse	08	15



Target				
SCENARIO	PRODUCT	PERIOD	QUANTITY	PRICE
2051BDG	Mouse	06	25	25
2051BDG	Mouse	07	35	25
2051BDG	Mouse	08	15	25

- Open new rows
- Enrich existing columns of a dataset
- **Conditions** fields are the key to perform the join
- **Attributes** fields are the information used to enrich the target

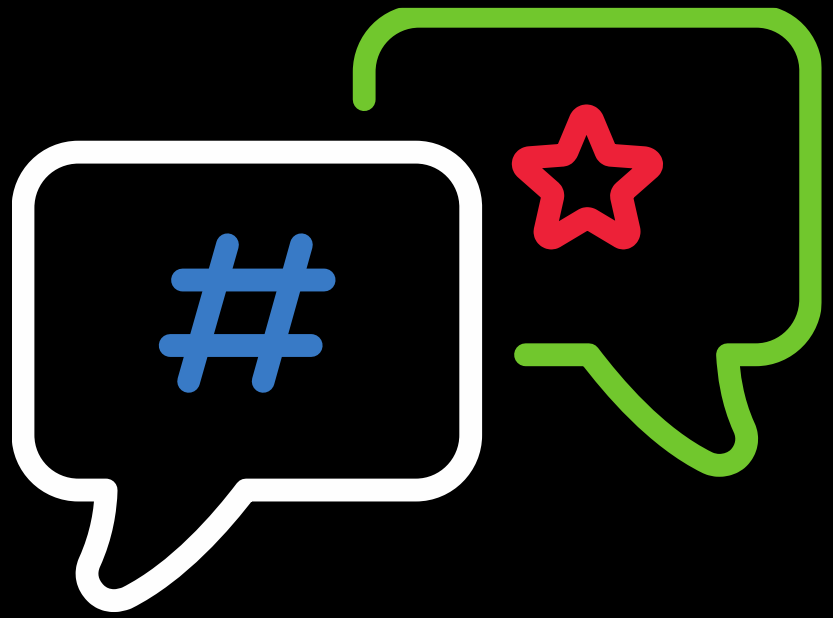
Now it’s your turn

Exercise # 7

Unpivoting with DTP operations



Let’s talk about: Enrichment with Formula: Recursive formulas (Revenue and discounted revenues calculation)



Revenue calculation: Activity steps

Target dataset: Revenue

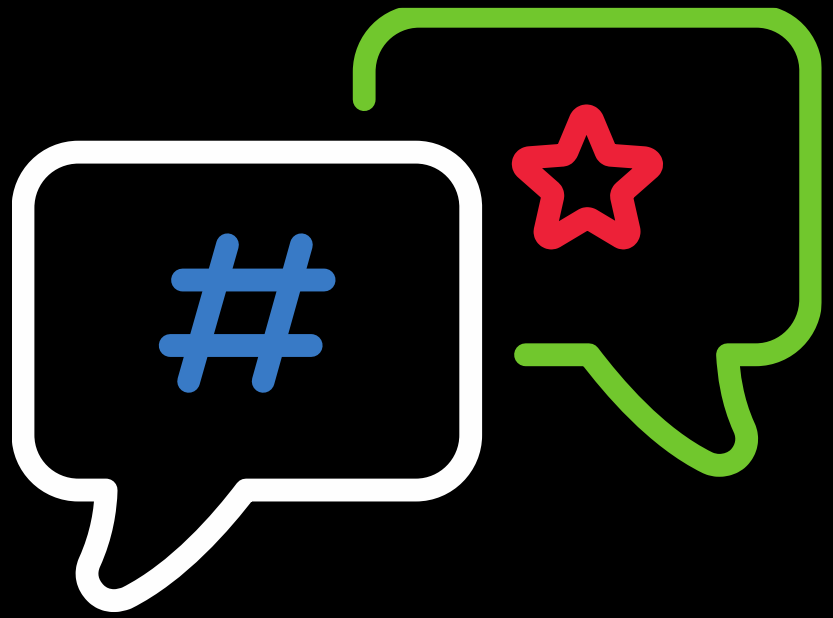
Insert data from **Unpivot Quantity**

Enrich with **Price** list

Enrich with **Discount** data

Calculate **Revenues and Discounted revenues**

Let’s talk about: Enrichment operations



Enrichment from another dataset

Source		
SCE (Condition)	PER (Condition)	PRICE (Attribute)
2050FCT	06	1000
2051BDG	12	2000



Target				
SCE	PER	PRICE	ENTITY	PRODUCT
2050FCT	06	1000		
2051BDG	12	2000		
2050BDG	03			

- Used to enrich existing target dataset column fields. Not to open new rows
- Attribute Source values are loaded in related Target field where condition columns match
- Condition combination must be univoque to avoid error

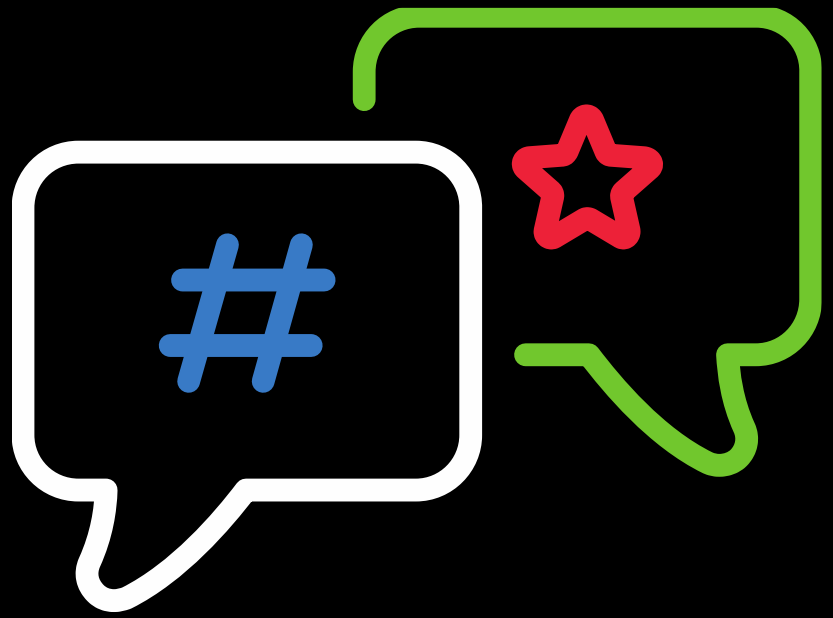
Enrichment from another dataset - Error

Source			
SCENARIO (Condition)	PRODUCT (Condition)	PERIOD (Attribute)	QUANTITY (Attribute)
2051BDG	Mouse	06	25
2051BDG	Mouse	07	35
2051BDG	Mouse	08	15

Target				
SCENARIO	PRODUCT	PERIOD	QUANTITY	PRICE
2051BDG	Mouse			20

- In Source dataset there are 3 rows with the same combination of conditions [**2051BDG - Mouse**]
- Enrichment is not allowed to generate new rows on target: **the system retrieve an error**

Let’s talk about: Revenues calculation: List of operations



Revenues calculation– List of operations

Source [Unpivot Quantity]					
Scenario	Entity	SKU	SKU Type	Period	Quantity
2053BDG	B00	S022	EL	01	10
2053BDG	B00	S005	SW	02	20
2053BDG	B00	S011	HW	12	120

Revenue calculation - Operations

- ✓ Insert from another dataset

Target [Revenue]									
Scenario	Entity	SKU	SKU Type	Period	Quantity	Price	Revenue	Discount	Discounted Revenue
2053BDG	B00	S022	EL	01	10				
2053BDG	B00	S005	SW	02	20				
2053BDG	B00	S011	HW	12	120				

Revenues calculation– List of operations

Source [Price]				
Scenario (Condition)	Entity (Condition)	SKU (Condition)	Period (Condition)	Price (Attribute)
2053BDG	B00	S022	01	30
2053BDG	B00	S005	02	10
2053BDG	B00	S011	12	20

Revenue calculation - Operations

- ✓ Insert from another dataset
- ✓ Enrichment from another dataset

Target [Revenue]									
Scenario	Entity	SKU	SKU Type	Period	Quantity	Price	Revenue	Discount	Discounted Revenue
2053BDG	B00	S022	EL	01	10	30			
2053BDG	B00	S005	SW	02	20	10			
2053BDG	B00	S011	HW	12	120	20			

Revenues calculation– List of operations

Source [Discount by SKU type]	
SKU Type (Condition)	Discount (Attribute)
EL	12%
SW	10%
HW	15%

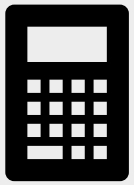


Revenue calculation - Operations

- ✓ Insert from another dataset
- ✓ Enrichment from another dataset
- ✓ Enrichment with reference dataset

Target [Revenue]									
Scenario	Entity	SKU	SKU Type	Period	Quantity	Price	Revenue	Discount	Discounted Revenue
2053BDG	B00	S022	EL	01	10	30		12%	
2053BDG	B00	S005	SW	02	20	10		10%	
2053BDG	B00	S011	HW	12	120	20		15%	

Revenues calculation– List of operations



1. **Revenue = Price X Quantity**
2. **Discounted Revenue = Revenue X (1-Discount)**



Revenue calculation - Operations

- ✓ Insert from another dataset
- ✓ Enrichment from another dataset
- ✓ Enrichment with reference dataset
- ✓ Enrichment with formula

Target [Revenue]

Scenario	Entity	SKU	SKU Type	Period	Quantity	Price	Revenue	Discount	Discounted Revenue
2053BDG	B00	S022	EL	01	10	30	300	12%	264
2053BDG	B00	S005	SW	02	20	10	200	10%	180
2053BDG	B00	S011	HW	12	120	20	2400	15%	2040

Enrichment with formula: Recursive formula

Enrichment with formula allows recursive formula among different matching fields:

- Reference can be done on target fields
- Reference is expressed with code of target field between @...@ = @REVENUE@

Enrichment with formula – matching fields

1) REVENUE = PRICE * QUANTITY
2) GROSS MARGIN = @REVENUE@ - COSTS

SKU	PRICE	QUANTITY	REVENUE	COSTS	GROSS MARGIN
S001	5	10	50	30	20
S002	10	20	200	100	100

Avoid writing multiple Enrichment with formula

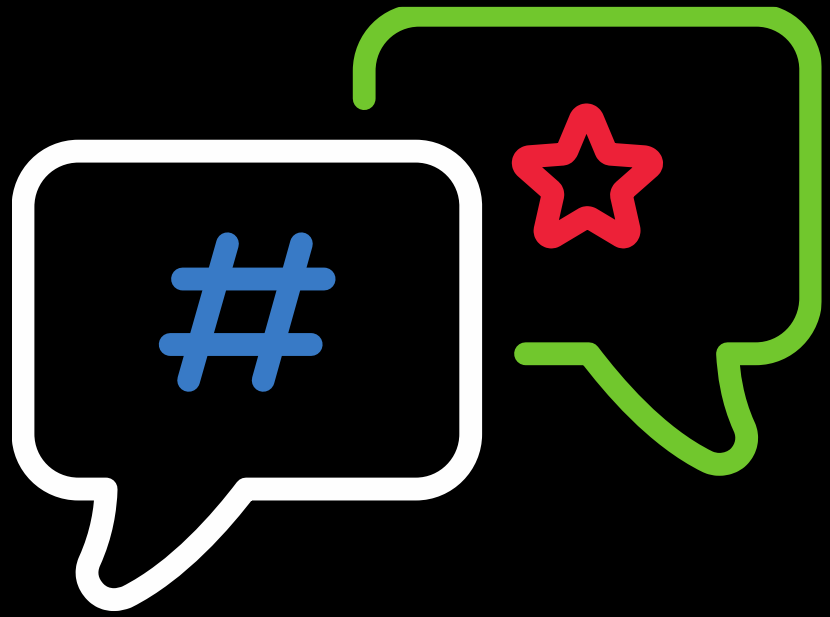
Now it’s your turn

Exercise # 8

Use of Recursive formula



Let’s talk about: Check operations



Coherence check

- Check if Target rows have the same combination of values of Source on the specified columns
- Warning message in case of inconsistencies

Source		
SCENARIO	PRODUCT	QUANTITY
2051BDG	Mouse	200
2051BDG	Laptop	350

Target			
SCENARIO	PRODUCT	QUANTITY	PRICE
2051BDG	Mouse		10
2051BDG	Laptop		30
2051BDG	Keyboard		45

⚠ **[Scenario: 2051BDG Product: Keyboard]** is a missing combination in Source dataset

Now it’s your turn

Exercise # 9

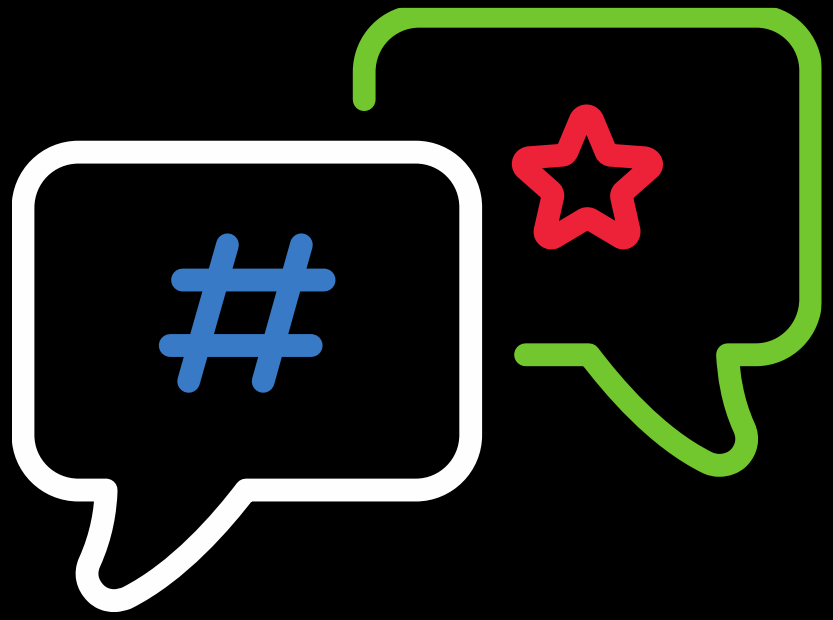
Check the consistency of data



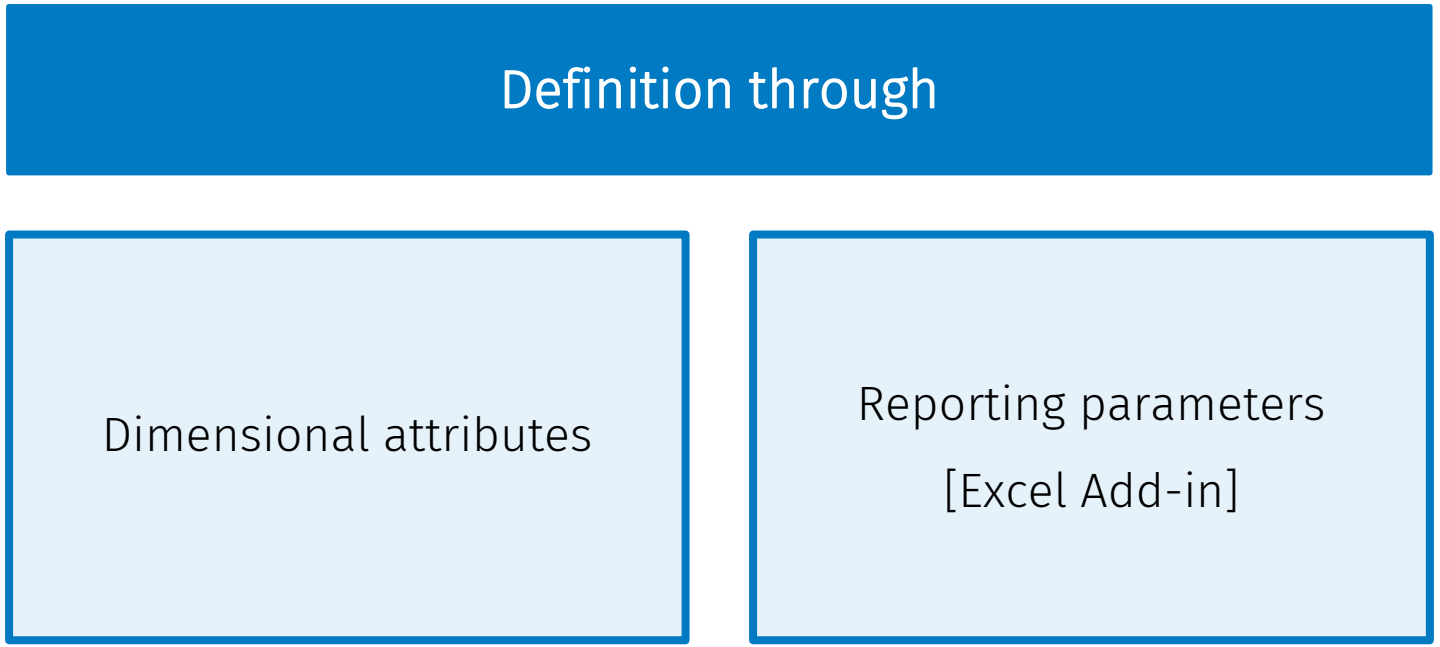
Recap Operations

Operation	Effect on target dataset
Loading to AW	Load data running Datasource mapping
Insert from another dataset	Insert new rows from source dataset
Enrichment with formula	Enrich target field with the result of a SQL formula
Enrichment from another dataset	Enrich existing rows with data from source dataset
Enrichment with reference dataset	Enrich existing rows as a lookup table. If reference dataset is a deploy one, it allows you to open new rows
Join from another dataset	Allows you to Enrich existing rows with data from source dataset and to open new rows
Coherence check	Check consistency of data

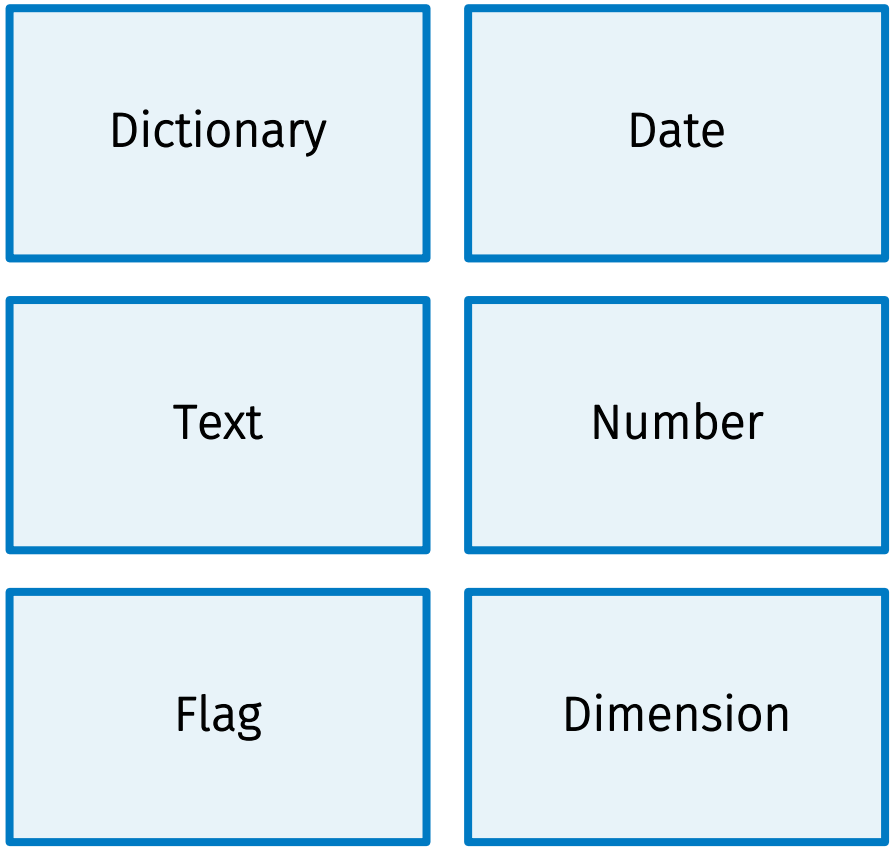
Let’s talk about: Dimensional and Reporting parameters



New types of parameters: definition



Reporting parameters: selection from



Parameters: where to use

Usable in

- Any DTP/SQL clause
  - Filters
  - Run conditions
  - Enrichment with formulas
  - Database Objects
  - Special methods
  - Datasources

⚠ Not available in queries of AW dictionary

Parameters recalling

Dimensional attribute

- ${$Entity.code}

Reporting parameter

- ${@PARAMETER}

Syntax examples

${$Entity.currency.code}

- Dimensional attribute retrieving entity currency

${@FLAG}='1'

- Flag parameter identity used in a DTP filter

${@DICTIONARY}

- Reporting parameter used to enrich SQL clause with elements selected from dictionary

${@$ANL_REGION_10_P.parent.code}

- Reporting parameter mixed with dimensional attribute retrieving the parent of an element selected from an analytical dimension hierarchy

DTP filters

Activity

- **Reporting layer filter:** preserves data from deletion
- **Run condition:** to execute the activity or not (It cannot be a condition based on a column of the dataset)

DTP filters

Operation

- Depending on operation type
- **Target filter**: filters data involved in the operation
- **Enrich only null values**
- **Source filter**: applied on the source table to retrieve data
- **Reference dataset filter:** applied on Reference dataset elements
- **Run Condition**

Old parameters	Old Syntax	How to replace
Workspace @...@	@PARAM@	Reporting Parameter
System #...#	#ENTITIES#	Attribute: ${$Entity.code}
	#SCENARIOS#	Attribute: ${$Scenario.code}
	#PERIODS#	Attribute: ${$Period.code}
	#TARGET_PARTITION#	Attribute: ${DATASETPARTITION(....)}
Special Method &...&	&TABLE1&	Attribute: ${DATASETPARTITION(....)}
	&STRING1&	Reporting parameter: Text

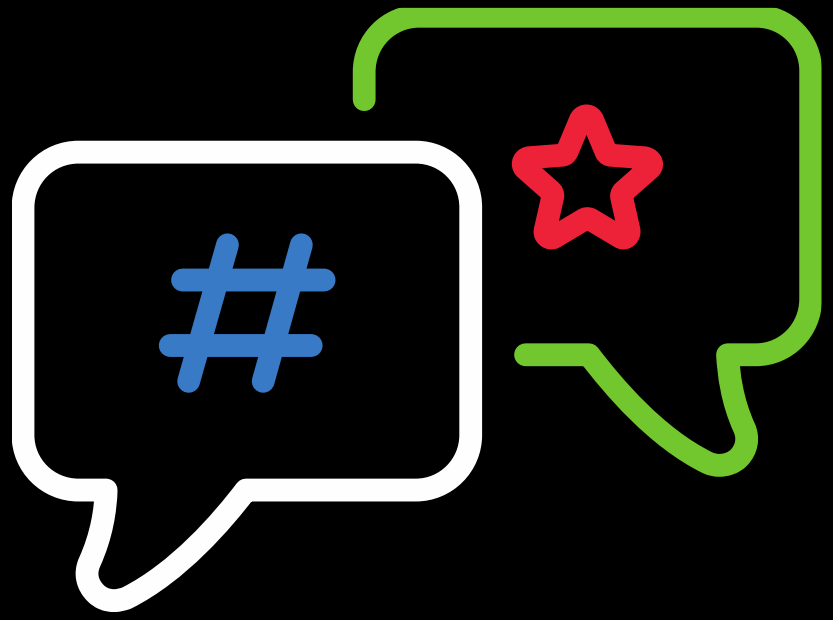
Now it’s your turn

Exercise # 10

Use of parameters



Let’s talk about: Virtual Datasets

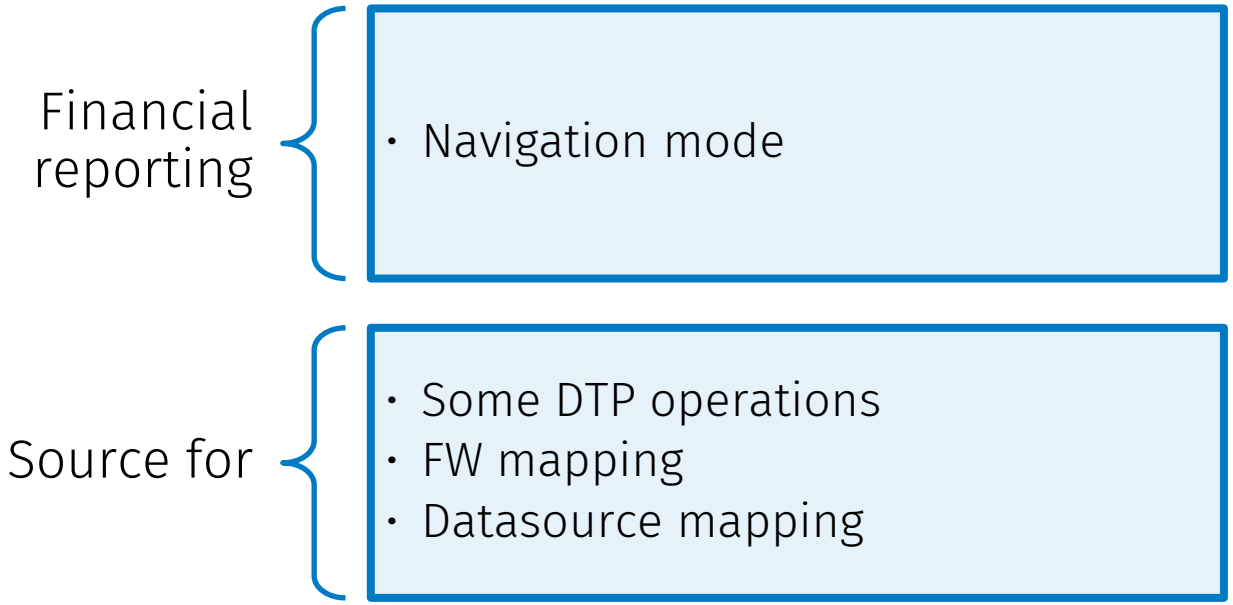


Virtual dataset

Main features

- Used to **combine records of one or more online physical datasets**
- It is a logical definition (**View**) based on the join among datasets fields
- **Dimensional fields:** fields of the dataset put in join
- **Formula fields** can be added
- Definition is permanent, data retrieved depends on the **moment of access**

Virtual dataset - Typical uses



Virtual Dataset - Example

Join defined on **Scenario, Period, Product**

Price

Scenario	Period	Product	Price
2057ACT	12	P01	500
2057ACT	12	P02	300
2057ACT	06	P03	100

+

Quantity

Scenario	Period	Product	Quantity
2057ACT	12	P01	10
2057ACT	12	P02	5
2057ACT	06	P09	30

Formula field

Virtual: Revenue

Scenario	Period	Product	Price	Quantity	Revenue ${Price} X ${Quantity}
2057ACT	12	P01	500	10	5000
2057ACT	12	P02	300	5	1500

Virtual dataset – Formula fields

Definition

Added manually

Syntax

Virtual dataset fields must be referred between **${}**
Example: **${PRICE} X ${QUANTITY}**

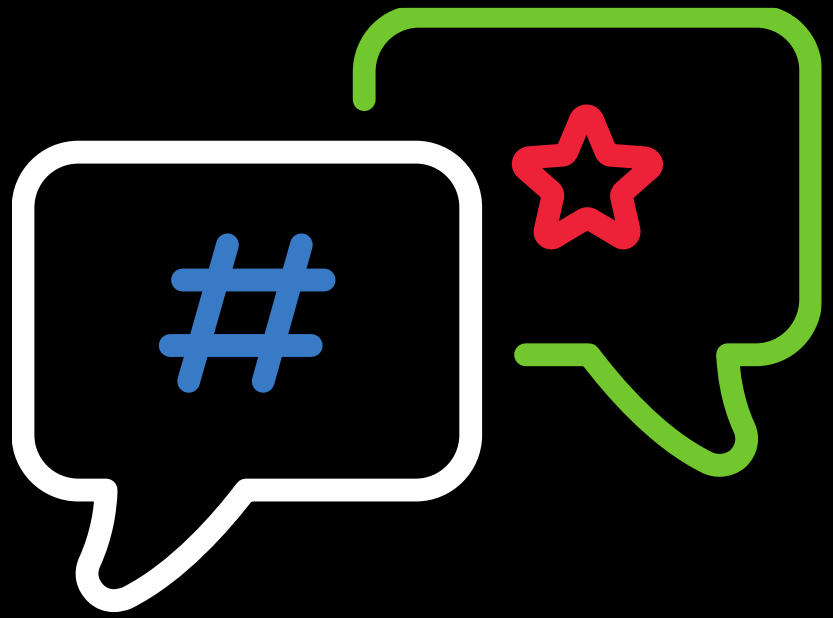
Now it’s your turn

Exercise # 11

Virtual dataset: an alternative way of calculating

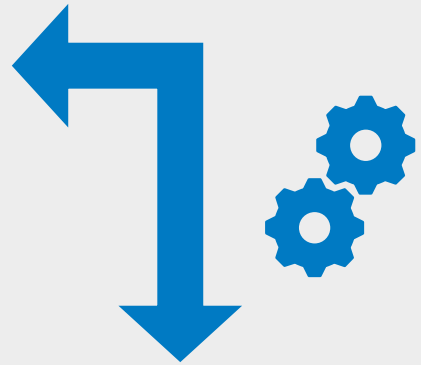


Let’s talk about: Unpivoting with Virtual Dataset (Optional)



Quantity and Unpivot Quantity

Quantity							
Scenario	Entity	SKU	SKU Type	P01	P02	....	P12
2053BDG	B00	S001	SW	10	20		120



Unpivot Quantity					
Scenario	Entity	SKU	SKU Type	Period	Quantity
2053BDG	B00	S001	SW	01	10
2053BDG	B00	S001	SW	02	20
2053BDG	B00	S001	SW	....	....
2053BDG	B00	S001	SW	12	120

Virtual dataset Unpivot Quantity

Join defined on **PERIOD**

Mapping period dataset

PERIOD	PERIOD_PIVOT
12	01
12	02
12	.....
12	12

+

Quantity

Scenario	PERIOD	Entity	SKU	SKU Type	P01	P02	....	P12
2053BDG	12	B00	S001	SW	10	20		120

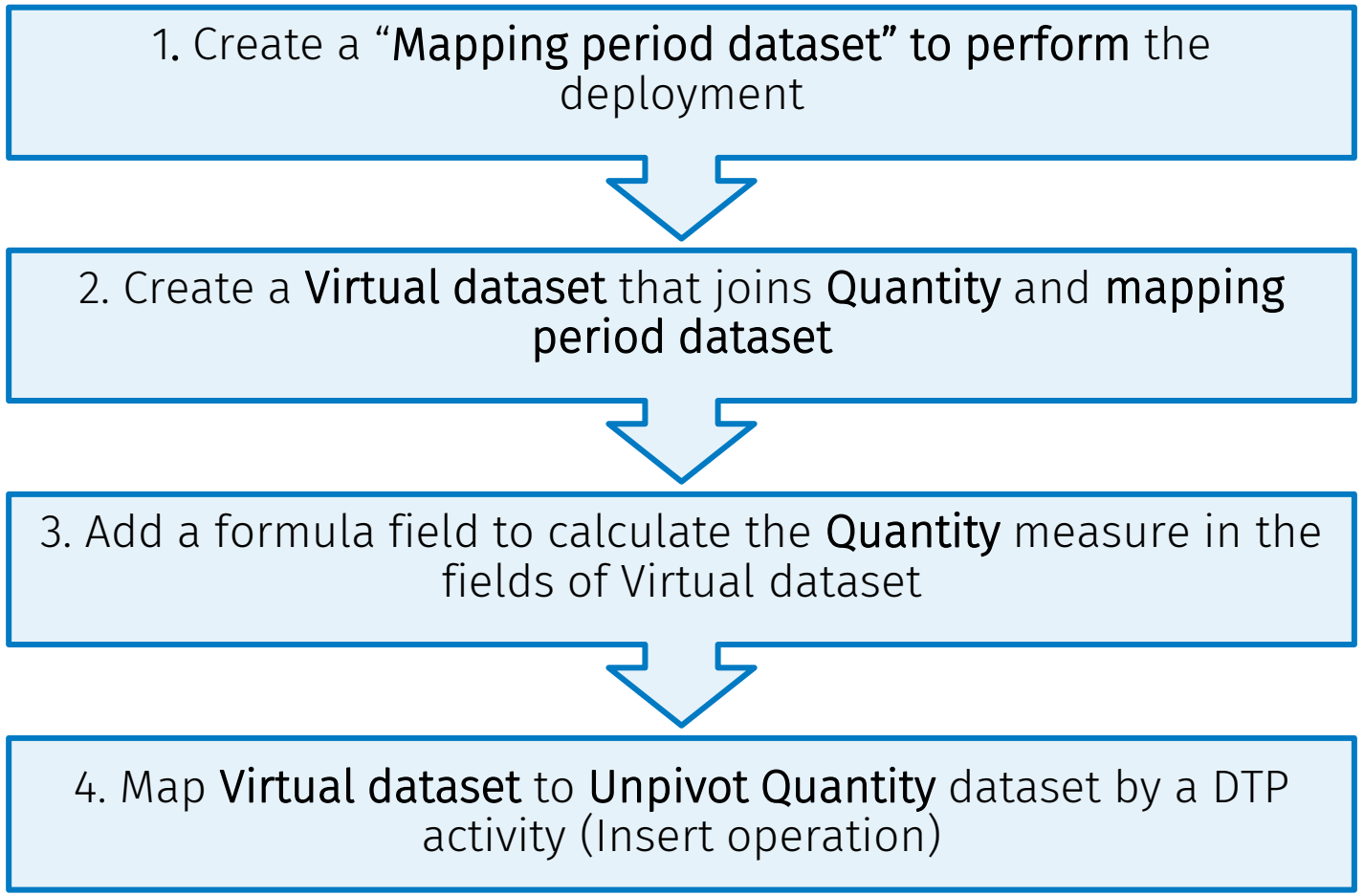
Formula field

```
CASE PERIOD_PIVOT
WHEN '01' THEN P01
WHEN '02' THEN P02
WHEN '03' THEN P03
.........
WHEN '12' THEN P12
ELSE 0 END
```

Virtual unpivot quantity

Scenario	Period	PERIOD_PIVOT	Entity	SKU	SKU Type	Quantity
2053BDG	12	01	B00	10	S001	10
2053BDG	12	02	B00	5	S001	20
2053BDG	12	.....				.....
2053BDG	12	12	B00		S001	120

Unpivoting with virtual dataset



Now it’s your turn

Exercise # 12

Unpivoting with Virtual dataset





Takeaway

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Leverage your social capital by connecting with other finance and operations professionals

Earn points up the leader board and redeem rewards

Here's the way the hub works

1 Log into the CCH Tagetik Community and click on the Insiders tab

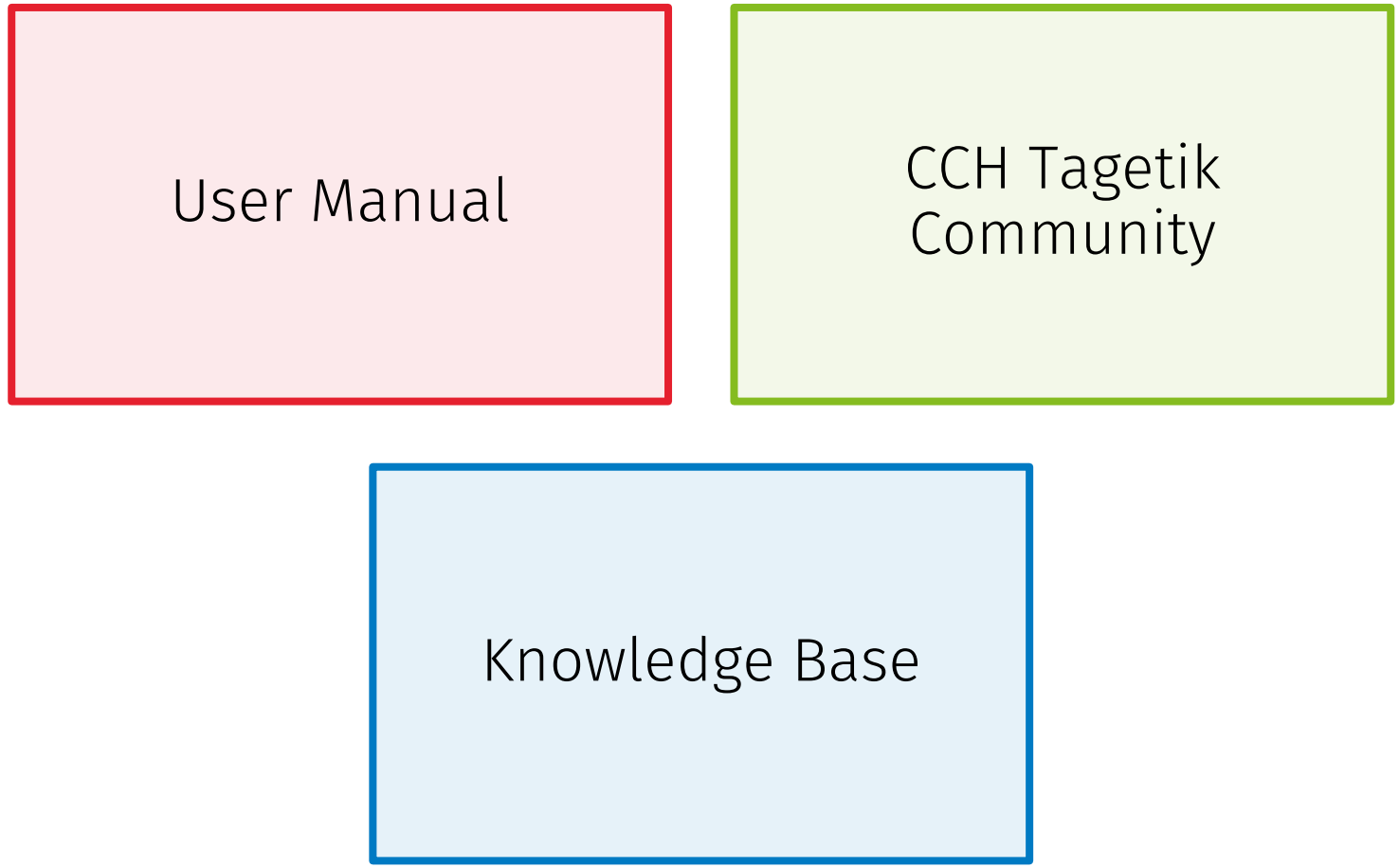
2 Poke around and complete challenges earning points to redeem rewards

3 Get recognized for all the support you give CCH Tagetik

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Next training and/or Assessment

Mail to info.academy@wolterskluwer.com

Thank you



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